

Components Explorer Layers

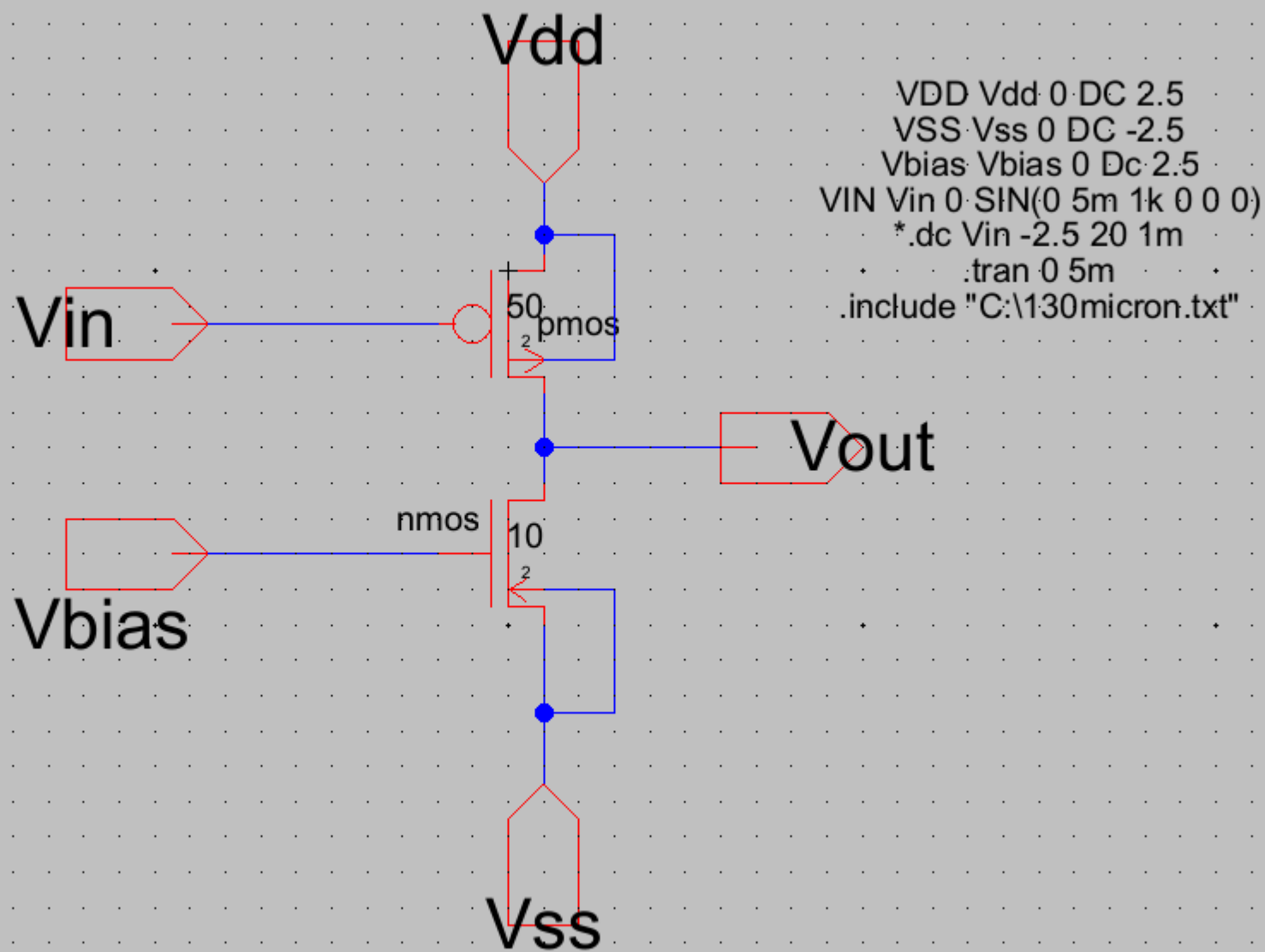
LIBRARIES

- 4zCommon_gate_amplifier1
 - AC.CG_amp{sch}
 - common_gate_amp{sch}
- 5zCS_amp1
 - CS_amp{sch}**
 - CS_amp_ac{sch}
- 7zdiff_amp1
 - Diff_amp_sim{sch}
- Diff_amplifier
 - Diff_amplifier{sch}
 - Diff_amplifier{ic}
- ac_diff_amp{sch}

spiceparts

ERRORS

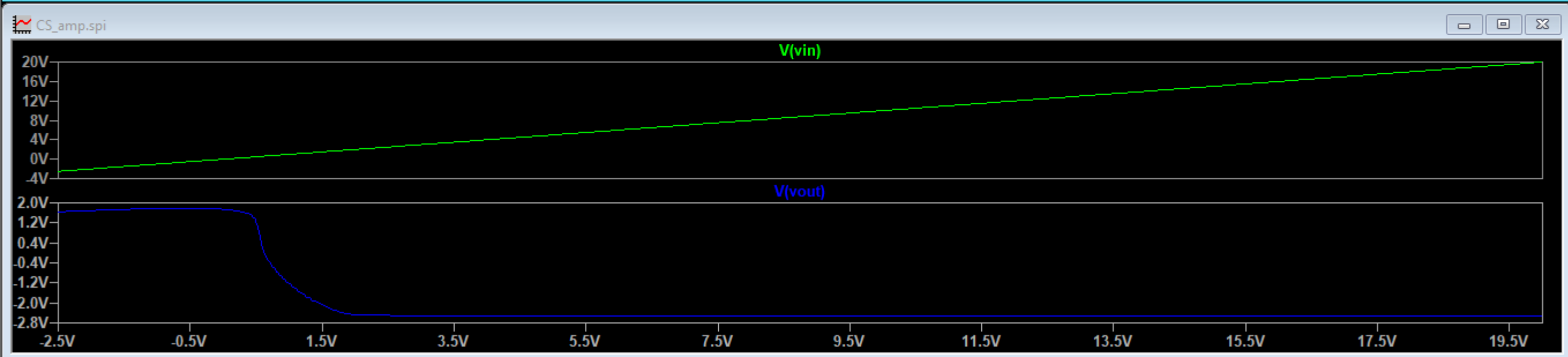
JOBS



```
CS_amp.spi

*** TOP LEVEL CELL: 5zCS_amp1:CS_amp{sch}
Mnmos-4@0 Vout Vbias Vss Vss nmos L=0.4U W=2U
Mpmos-4@0 vdd Vin Vout vdd pmos L=0.4U W=10U

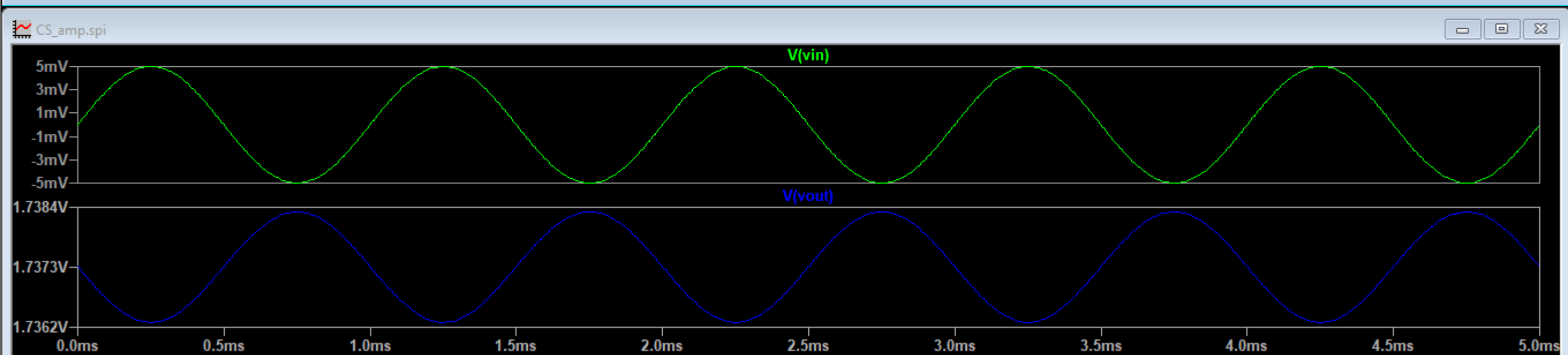
* Spice Code nodes in cell cell '5zCS_amp1:CS_amp{sch}'
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
Vbias Vbias 0 DC 2.5
VIN Vin 0 SIN(0 5m 1k 0 0 0)
.dc Vin -2.5 20 1m
*.tran 0 5m
.include "C:\130micron.txt"
.END
```

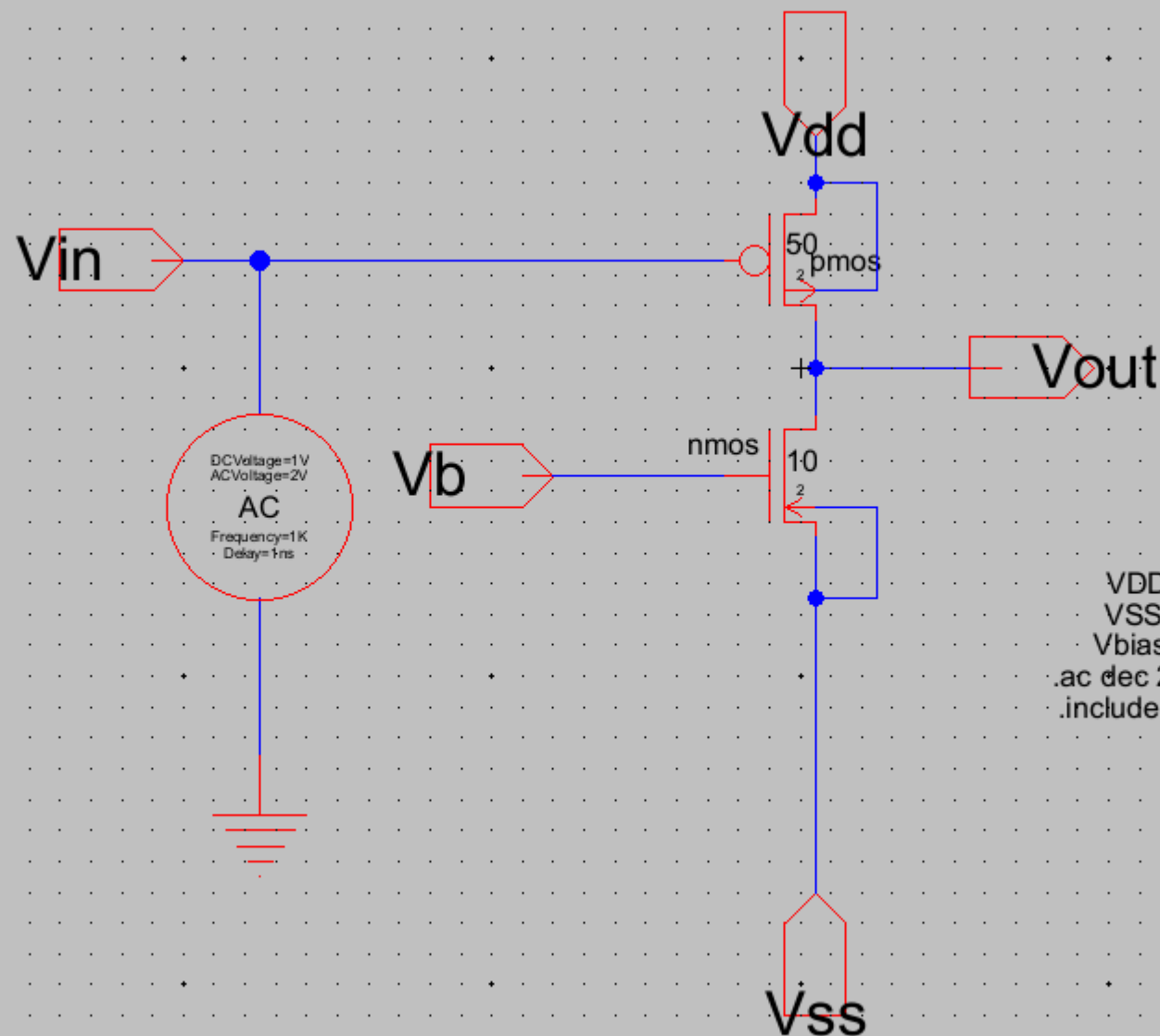
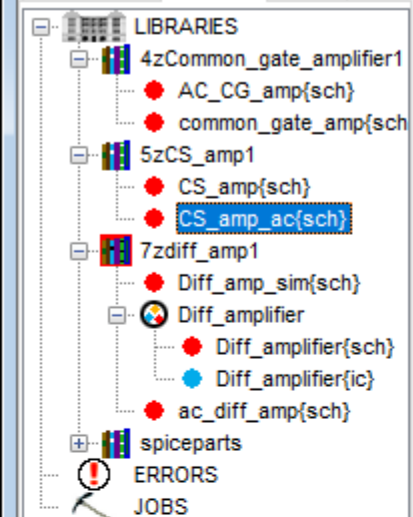


```
CS_amp.spi

*** TOP LEVEL CELL: 5zCS_amp1:CS_amp{sch}
Mnmos-4@0 Vout Vbias Vss Vss nmos L=0.4U W=2U
Mpmos-4@0 vdd Vin Vout vdd pmos L=0.4U W=10U

* Spice Code nodes in cell cell '5zCS_amp1:CS_amp{sch}'
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
Vbias Vbias 0 DC 2.5
VIN Vin 0 SIN(0 5m 1k 0 0 0)
*.dc Vin -2.5 20 1m
.tran 0 5m
.include "C:\130micron.txt"
.END
```





```
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
Vbias Vbias 0 DC 2.5
.ac dec 20 100 500000meg
.include "C:\130micron.txt"
```



CS_amp_ac.spi

```
.global gnd

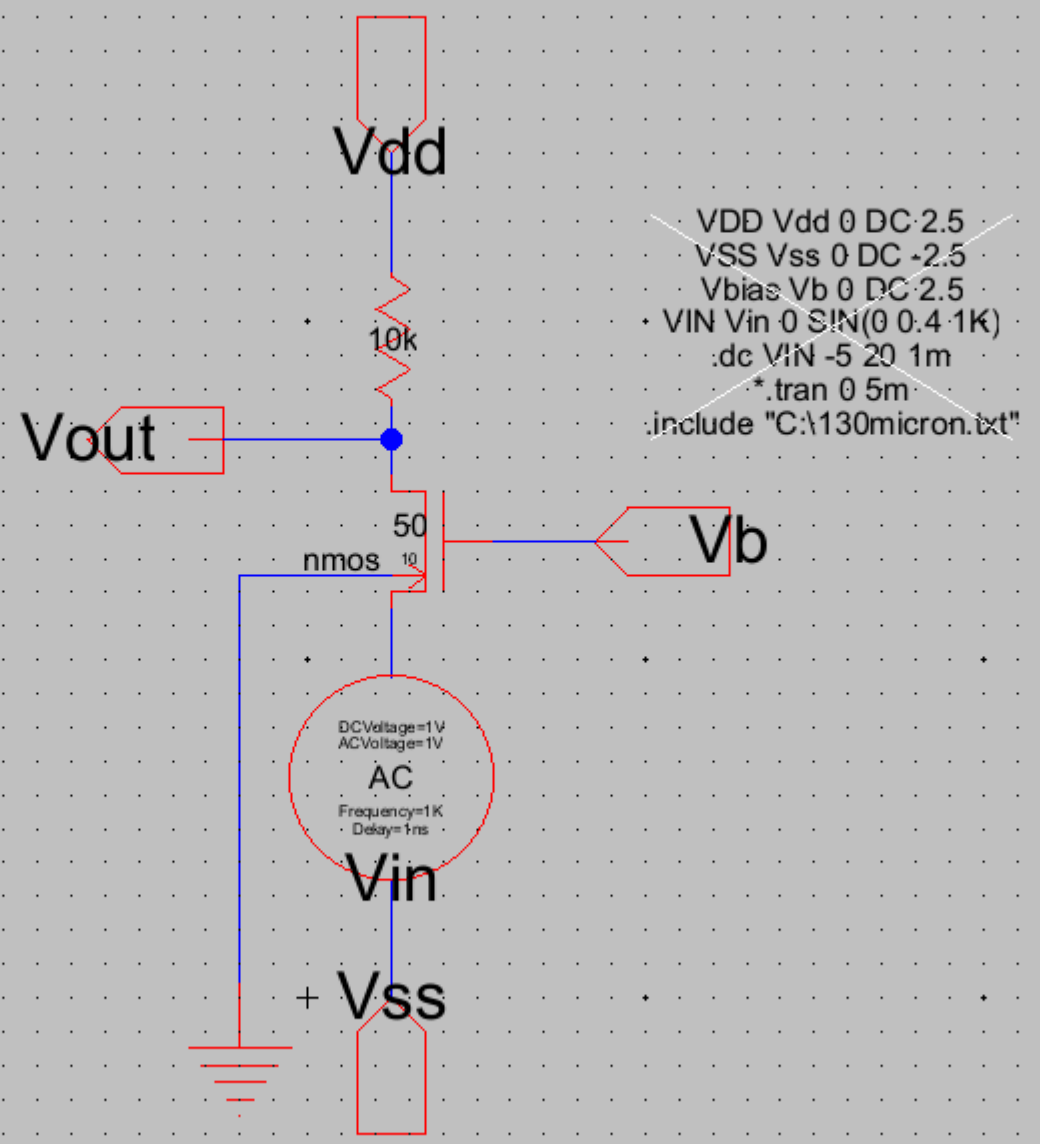
*** TOP LEVEL CELL: 5zCS_amp1:CS_amp_ac{sch}
Mnmos-4@0 Vout Vb Vss Vss nmos L=0.4U W=2U
Mpmos-4@0 vdd Vin Vout vdd pmos L=0.4U W=10U
VACVtag@0 Vin gnd DC=1V AC=2V SIN (1V 2V 1,000 1ns)

* Spice Code nodes in cell cell '5zCS_amp1:CS_amp_ac{sch}'
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
Vbias Vbias 0 Dc 2.5
.ac dec 20 100 500000meg
.include "C:\130micron.txt"
.END
```

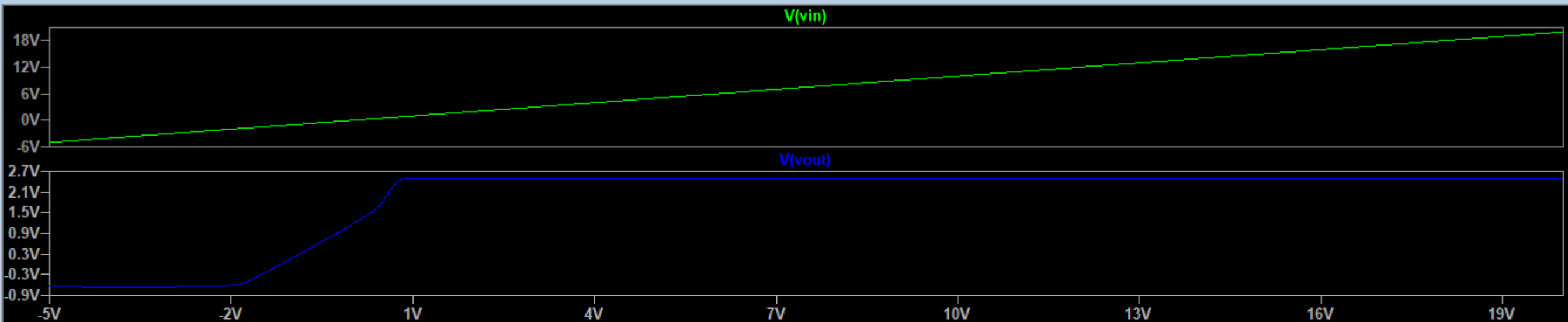
Components Explorer Layers

LIBRARIES

- 4zCommon_gate_amplifier1
 - AC.CG_amp{sch}
 - common_gate_amp{sch}
- 5zCS_amp1
 - CS_amp{sch}
 - CS_amp_ac{sch}
- 7zdiff_amp1
- spiceparts
- ERRORS
- JOBS



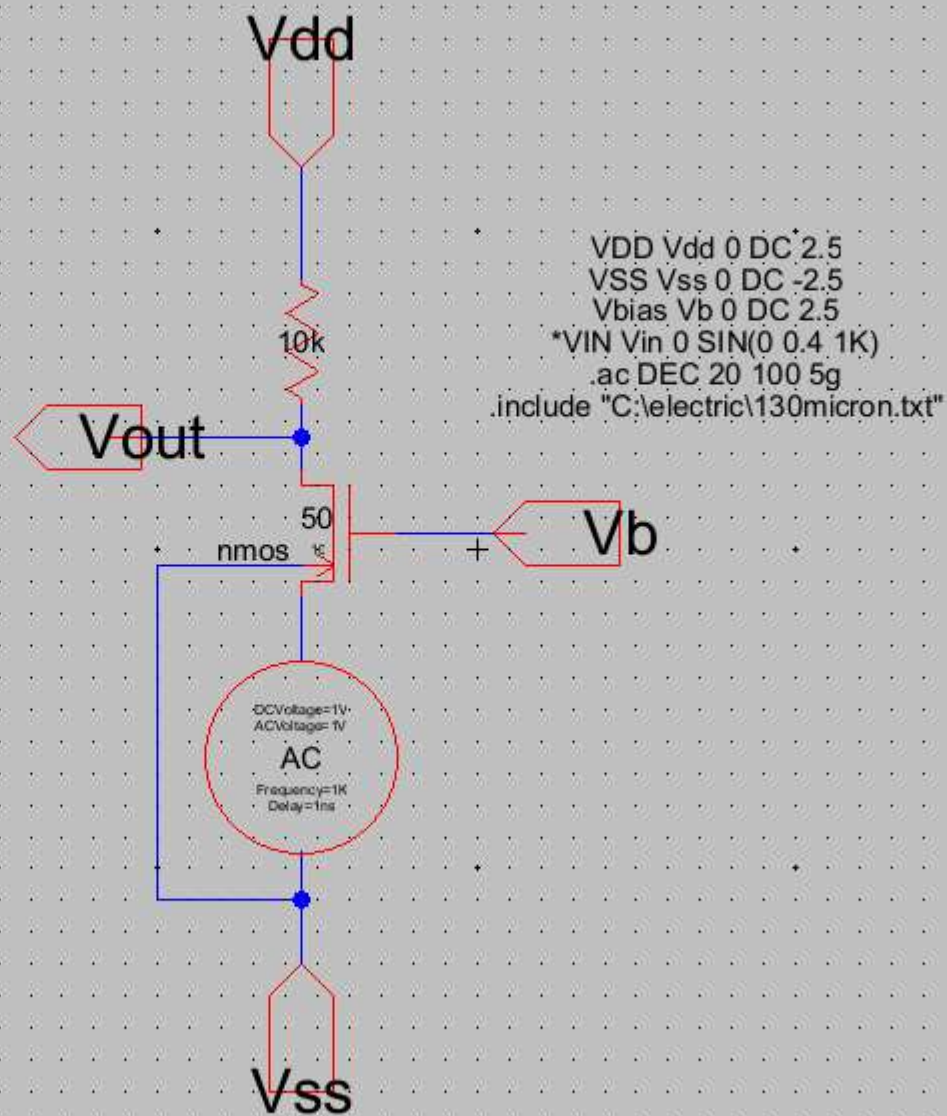
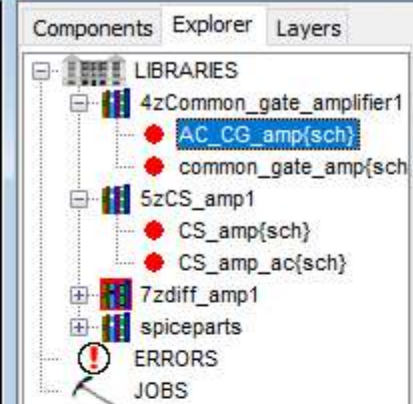
common_gate_amp.spi

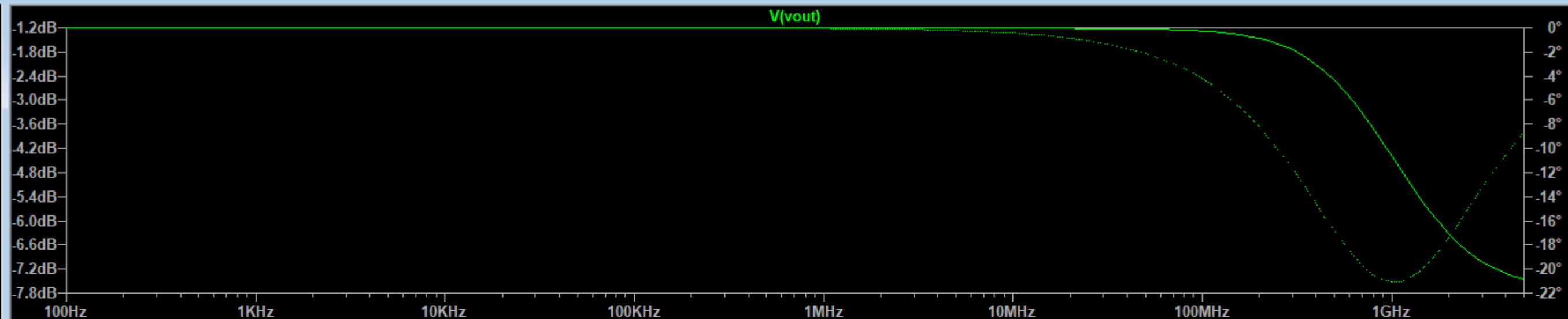


common_gate_amp.spi

```
Mnmos-4@0 Vout Vb net@1 gnd nmos L=2U W=10U
Rres@0 vdd Vout 10000
VACVtag@1 net@1 Vin DC=1V AC=1V SIN (1V 1V 1,000 1ns)
```

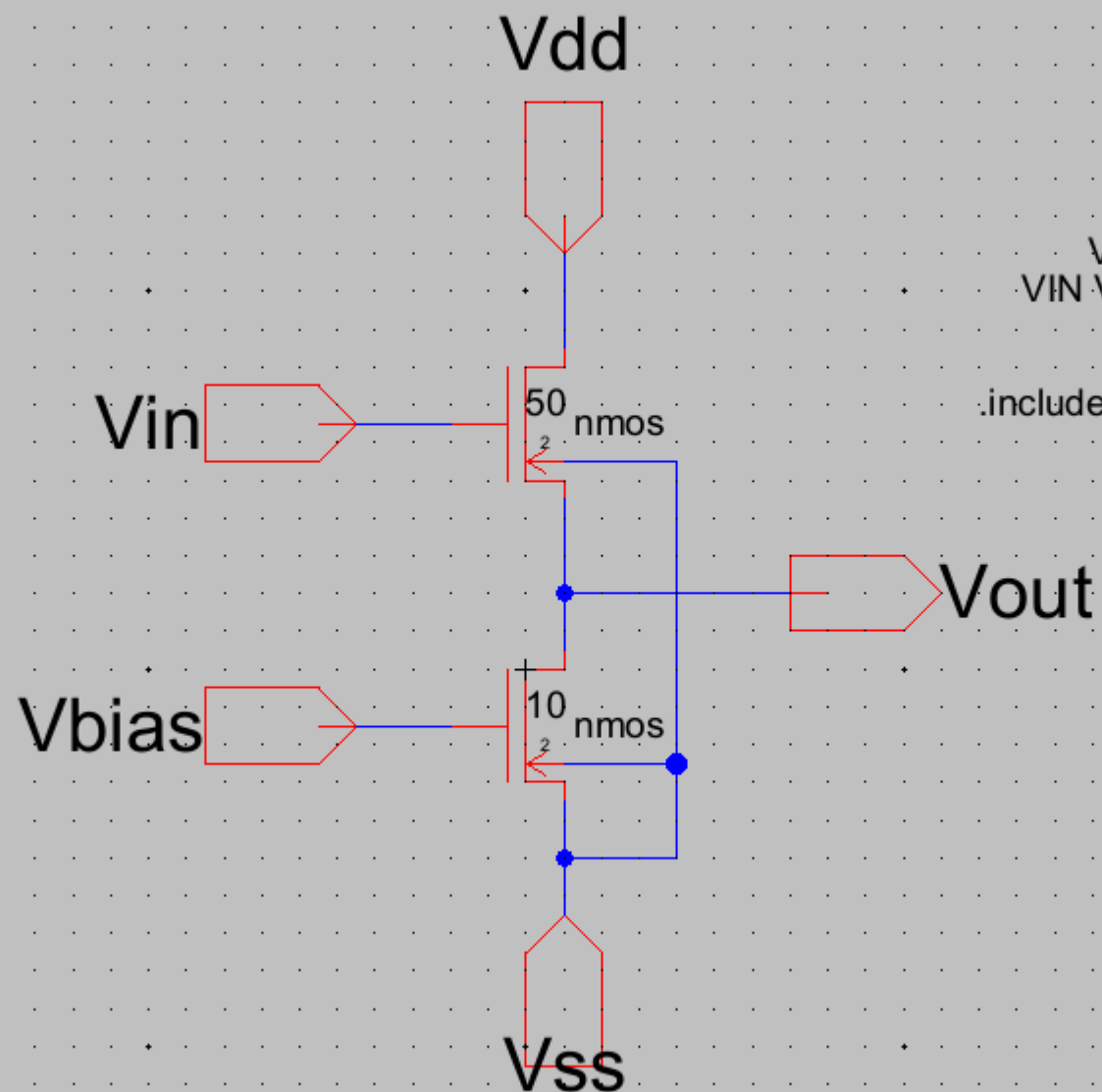
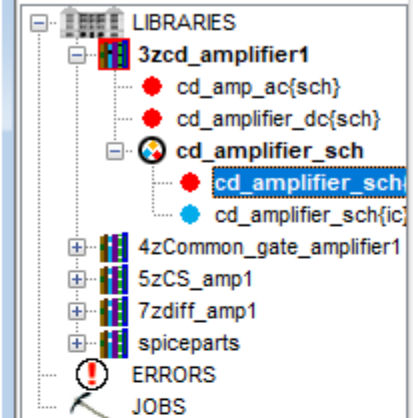
```
* Spice Code nodes in cell cell
*'4zCommon_gate_amplifier1:common_gate_amp{sch}'
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
Vbias Vb 0 DC 2.5
VIN Vin 0 SIN(0 0.4 1K)
.dc VIN -5 20 1m
*.tran 0 5m
.include "C:\130micron.txt"
.END
```



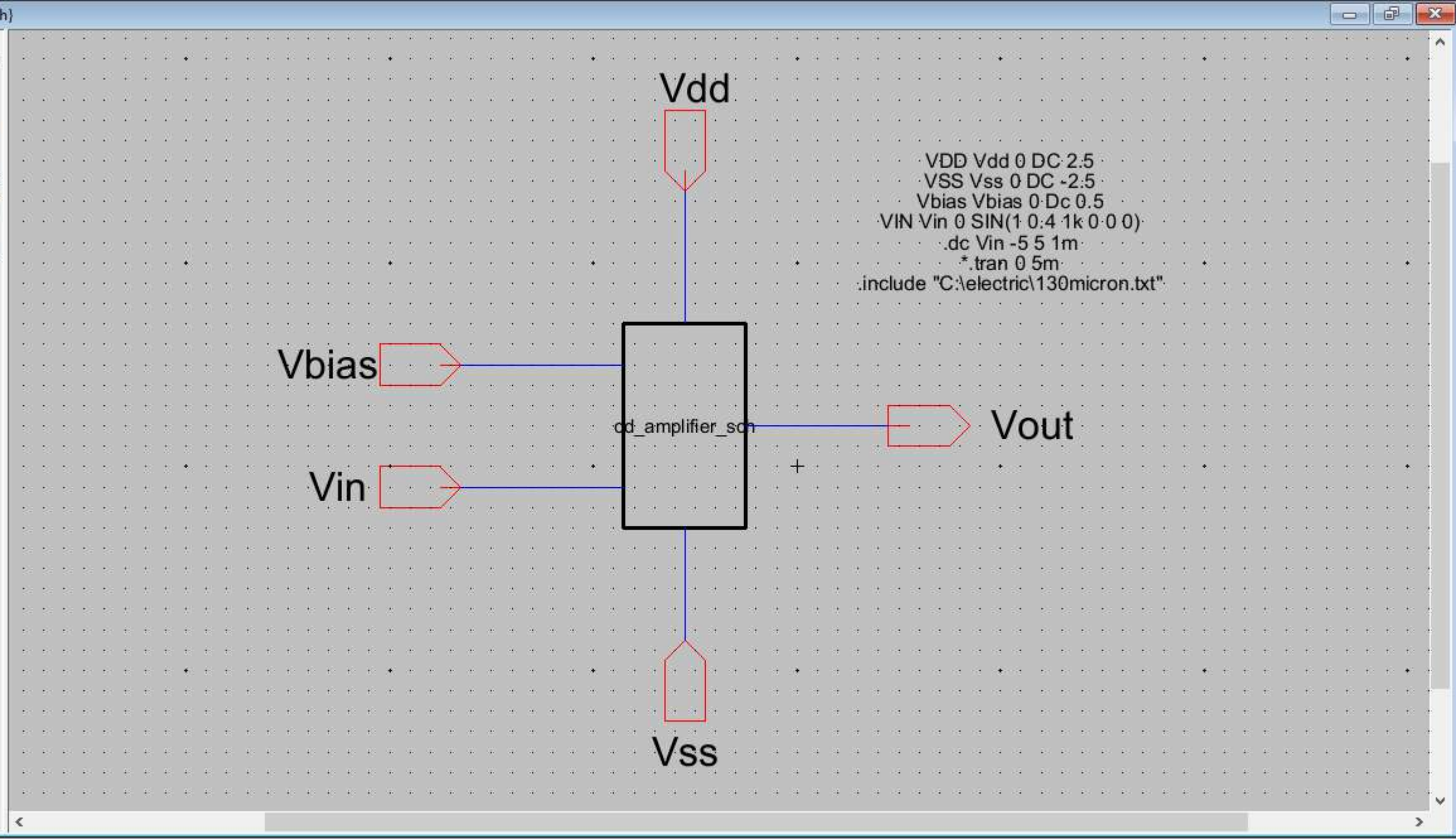
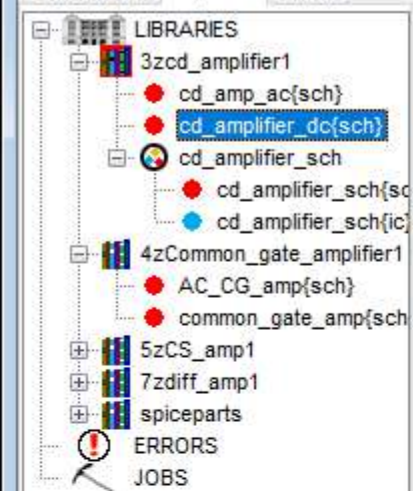


```
*** TOP LEVEL CELL: 4zCommon_gate_amplifier1:AC.CG_amp{sch}
Mnmos-4@0 Vout Vb net@0 Vss nmos L=2U W=10U
Rres@0 vdd Vout 10000
VACVtag@0 net@0 Vss DC=1V AC=1V SIN (1V 1V 1,000 1ns)

* Spice Code nodes in cell cell '4zCommon_gate_amplifier1:AC.CG_amp{sch}'
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
Vbias Vb 0 DC 2.5
*VIN Vin 0 SIN(0 0.4 1K)
.ac DEC 20 100 5g
.include "C:\130micron.txt"
.END
```

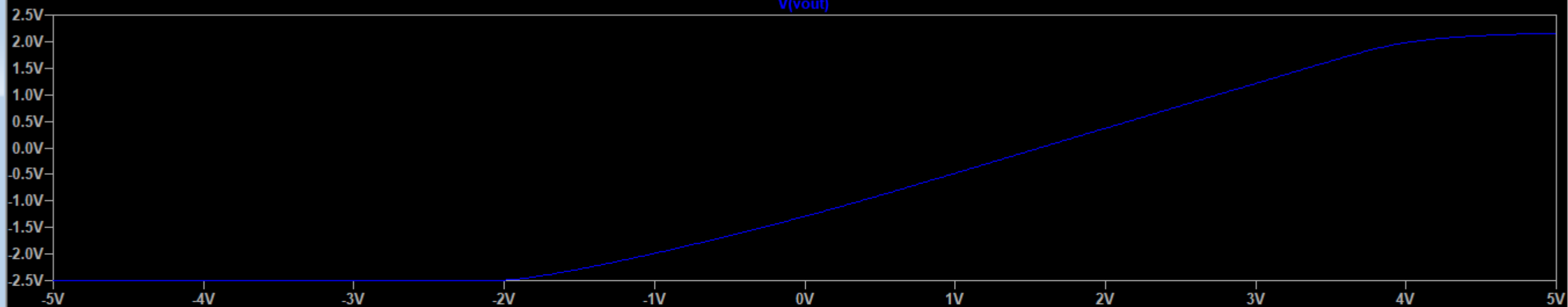


```
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
Vbias Vbias 0 DC 0.5
VIN Vin 0 SIN(1 0.4 1k 0 0 0)
.dc Vin -5 5 1m
*.tran 0 5m
.include "C:\electric\130micron.txt"
```



cd_amplifier_dc.spi

V(vout)



cd_amplifier_dc.spi



```
.ENDS cd_amplifier_sch
```

```
*** TOP LEVEL CELL: cd_amplifier_dc{sch}
```

```
Xcd_ampli@0 Vbias Vdd Vin Vout Vss cd_amplifier_sch
```

```
* Spice Code nodes in cell cell 'cd_amplifier_dc{sch}'
```

```
VDD Vdd 0 DC 2.5
```

```
VSS Vss 0 DC -2.5
```

```
Vbias Vbias 0 DC 0.5
```

```
VIN Vin 0 SIN(1 0.4 1k 0 0 0)
```

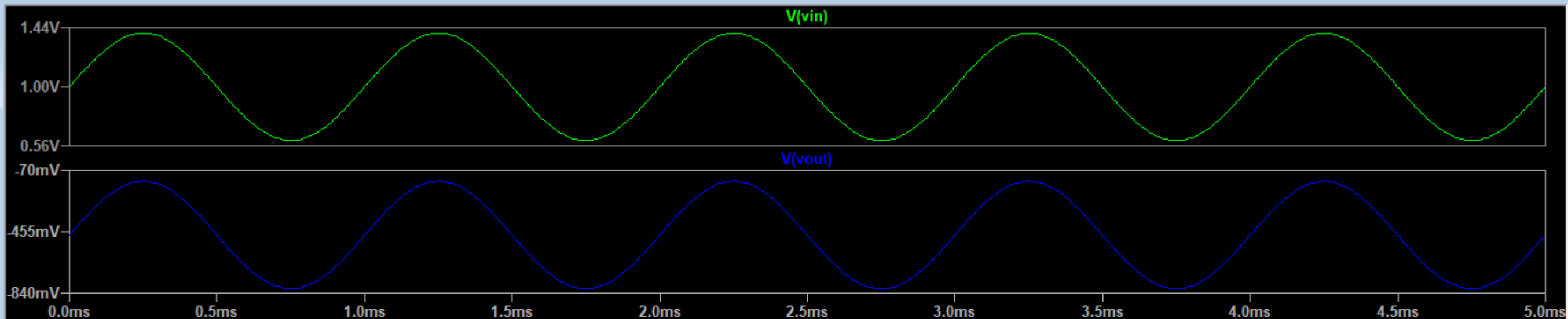
```
.dc Vin -5 5 1m
```

```
*.tran 0 5m
```

```
.include "C:\130micron.txt"
```

```
.END
```

cd_amplifier_dc.spi



cd_amplifier_dc.spi

```
.ENDS cd_amplifier_sch
```

```
*** TOP LEVEL CELL: cd_amplifier_dc{sch}
```

```
Xcd_ampli@0 Vbias Vdd Vin Vout Vss cd_amplifier_sch
```

```
* Spice Code nodes in cell cell 'cd_amplifier_dc{sch}'
```

```
VDD Vdd 0 DC 2.5
```

```
VSS Vss 0 DC -2.5
```

```
Vbias Vbias 0 DC 0.5
```

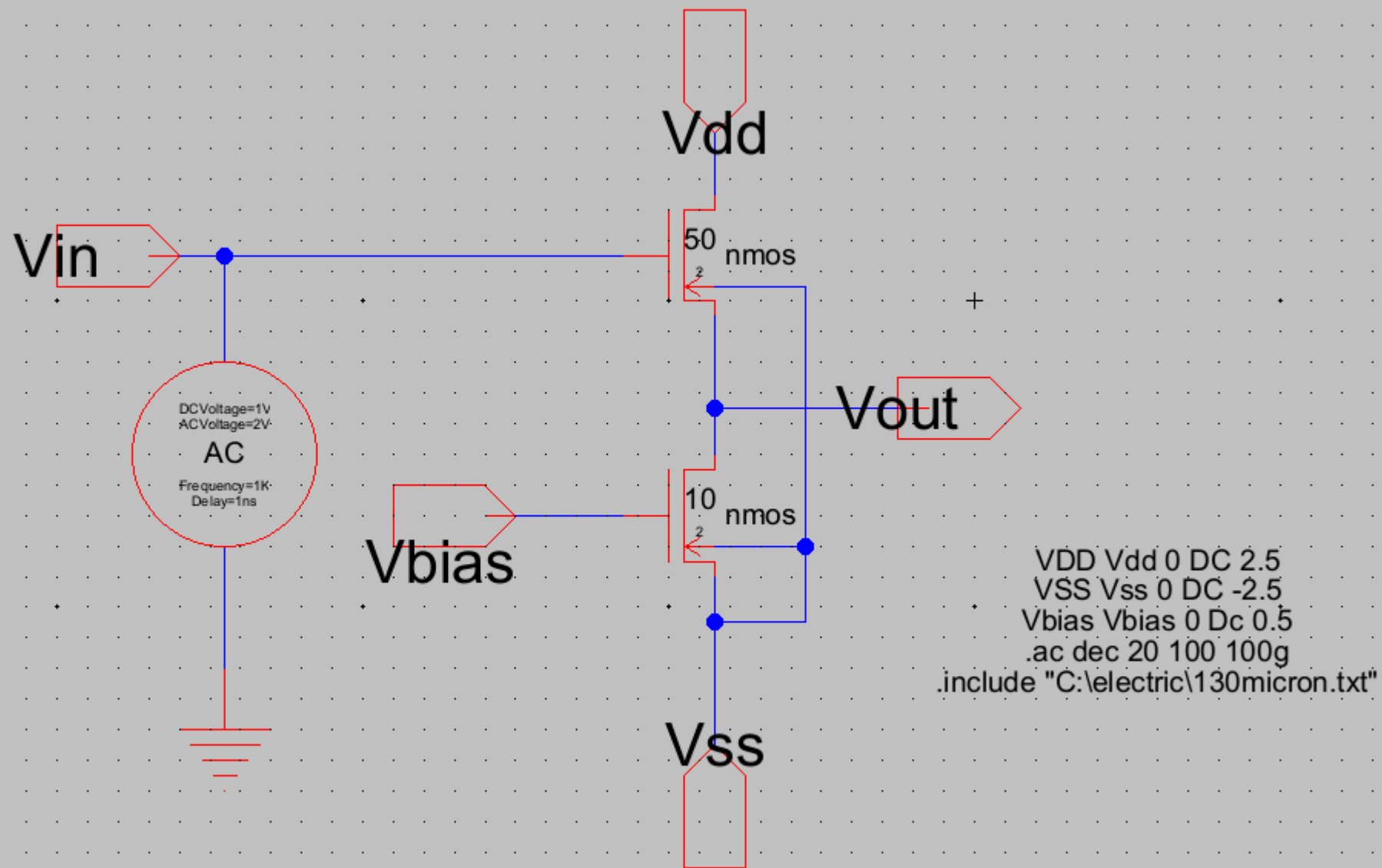
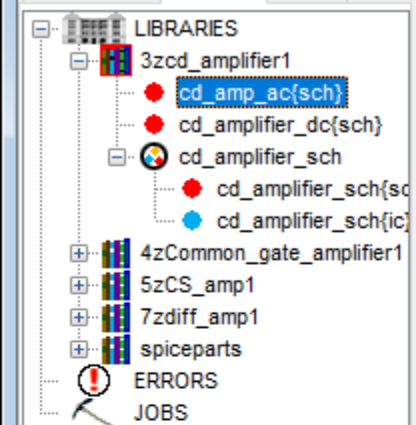
```
VIN Vin 0 SIN(1 0.4 1k 0 0 0)
```

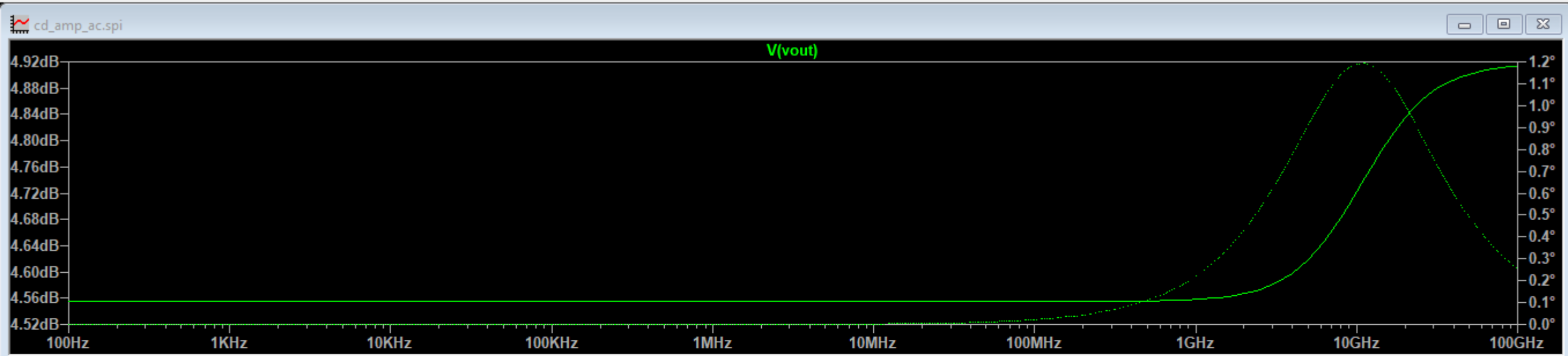
```
*|.dc Vin -5 5 1m
```

```
.tran 0 5m
```

```
.include "C:\130micron.txt"
```

```
.END
```



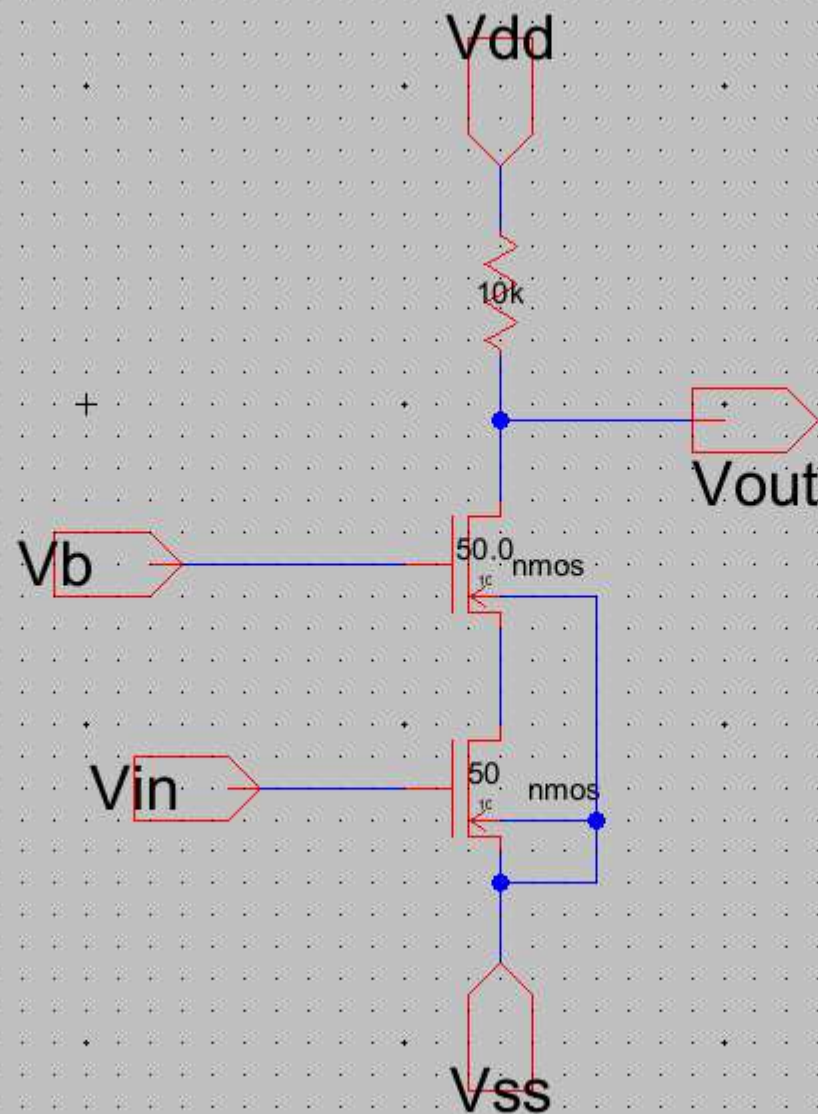
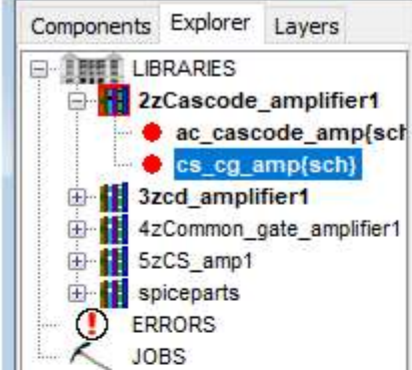


cd_amp_ac.spi

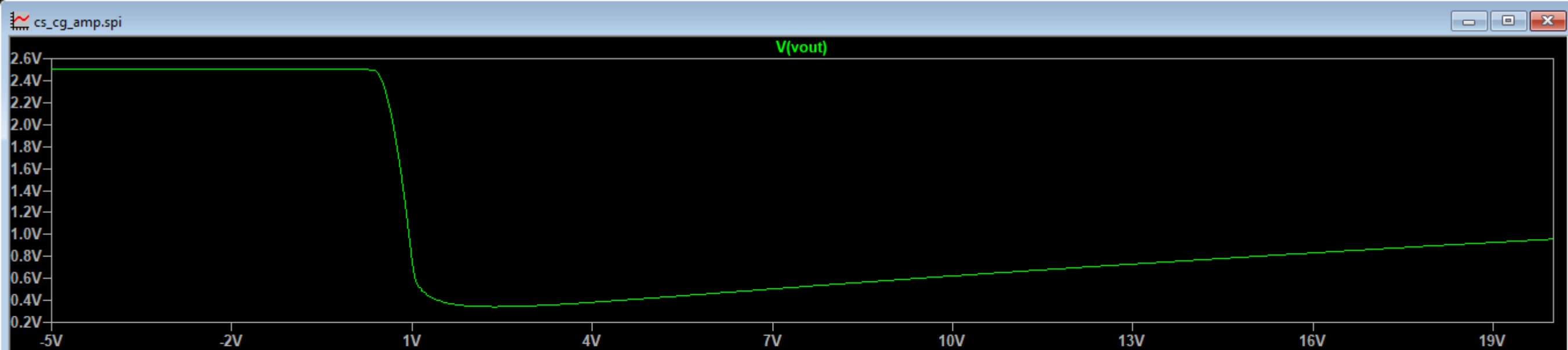
```
.global gnd

*** TOP LEVEL CELL: cd_amp_ac{sch}
Mnmos-4@0 vdd Vin Vout Vss nmos L=0.4U W=10U
Mnmos-4@1 Vout Vbias Vss Vss nmos L=0.4U W=2U
VACVtag@1 Vin gnd DC=1V AC=2V SIN (1V 2V 1,000 1ns)

* Spice Code nodes in cell cell 'cd_amp_ac{sch}'
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
Vbias Vbias 0 Dc 0.5
.ac dec 20 100 100g
.include "C:\130micron.txt"
.END
```



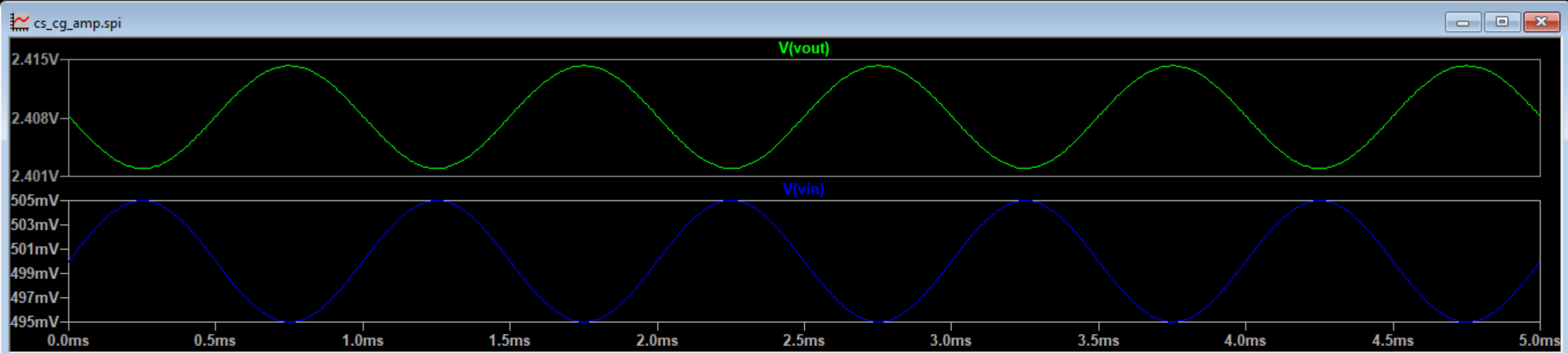
```
VDD Vdd 0 DC 2.5
VBias Vb 0 DC 2.5
VSS Vss 0 DC 0
VIN Vin 0 SIN(0.5 5m 1k 0 0 0)
*.dc VIN -5 20 1m
.tran 0 5m
.include "C:\130micron.txt"
```

cs_cg_amp.spi

```
*** TOP LEVEL CELL: cs_cg_amp{sch}
Mnmos-4@2 Vout Vb net@93 Vss nmos L=2U W=10U
Mnmos-4@3 net@93 Vin Vss Vss nmos L=2U W=10U
Rres@1 vdd Vout 10000

* Spice Code nodes in cell cell 'cs_cg_amp{sch}'
VDD Vdd 0 DC 2.5
VBias Vb 0 DC 2.5
VSS Vss 0 DC 0
VIN Vin 0 SIN(0.5 5m 1k 0 0 0)
.dc VIN -5 20 1m
*.tran 0 5m
.include "C:\130micron.txt"
.END
```



cs_cg_amp.spi

```
*** TOP LEVEL CELL: cs_cg_amp{sch}
Mnmos-4@2 Vout Vb net@93 Vss nmos L=2U W=10U
Mnmos-4@3 net@93 Vin Vss Vss nmos L=2U W=10U
Rres@1 vdd Vout 10000

* Spice Code nodes in cell cell 'cs_cg_amp{sch}'
VDD Vdd 0 DC 2.5
VBias Vb 0 DC 2.5
VSS Vss 0 DC 0
VIN Vin 0 SIN(0.5 5m 1k 0 0 0)
*.dc VIN -5 20 1m
.tran 0 5m
.include "C:\130micron.txt"
.END
```

Components Explorer Layers

LIBRARIES

2zCascode_amplifier1

• ac_cascode_amp{sch}

• cs_cg_amp{sch}

3zcd_amplifier1

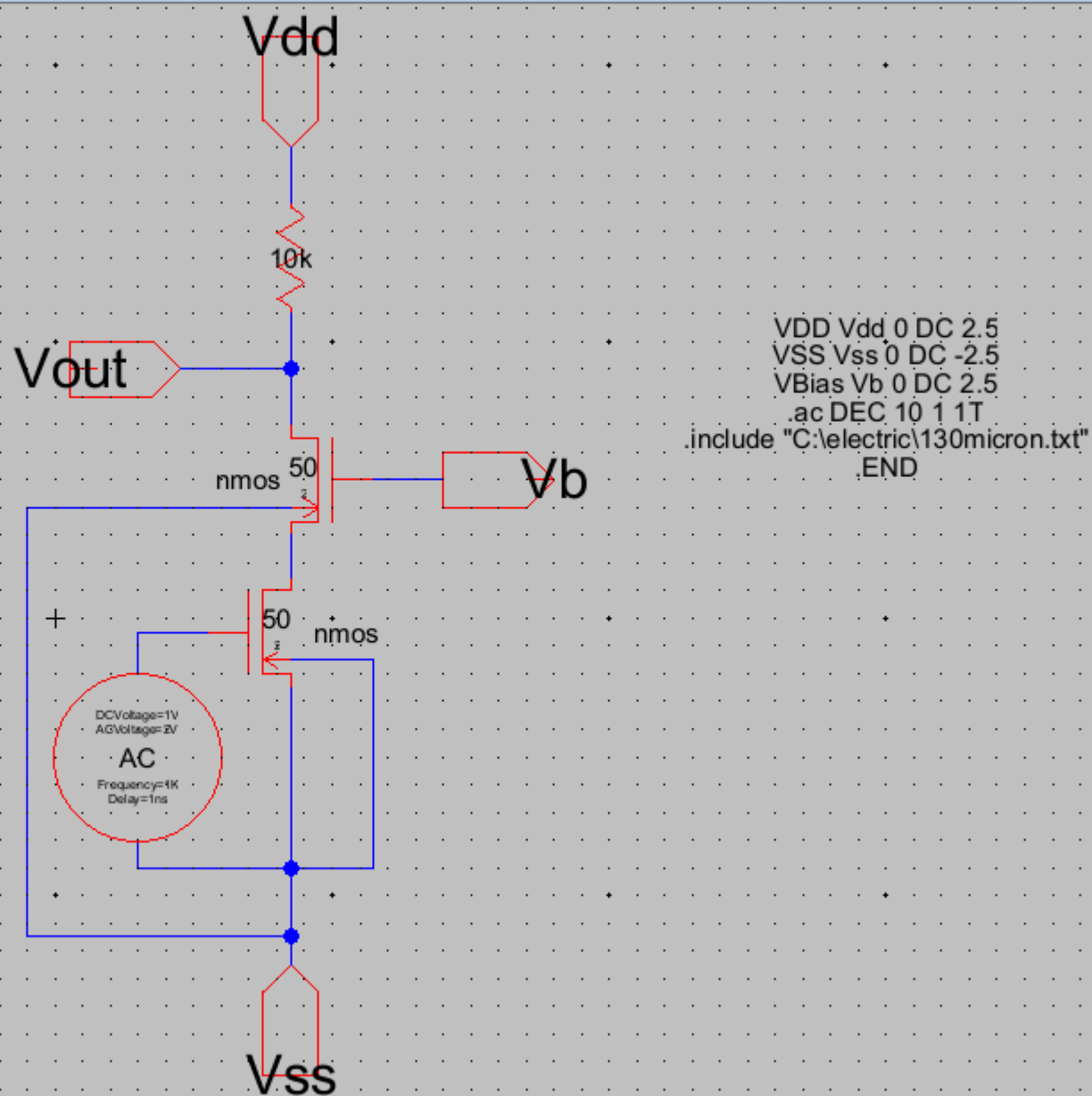
4zCommon_gate_amplifier1

5zCS_amp1

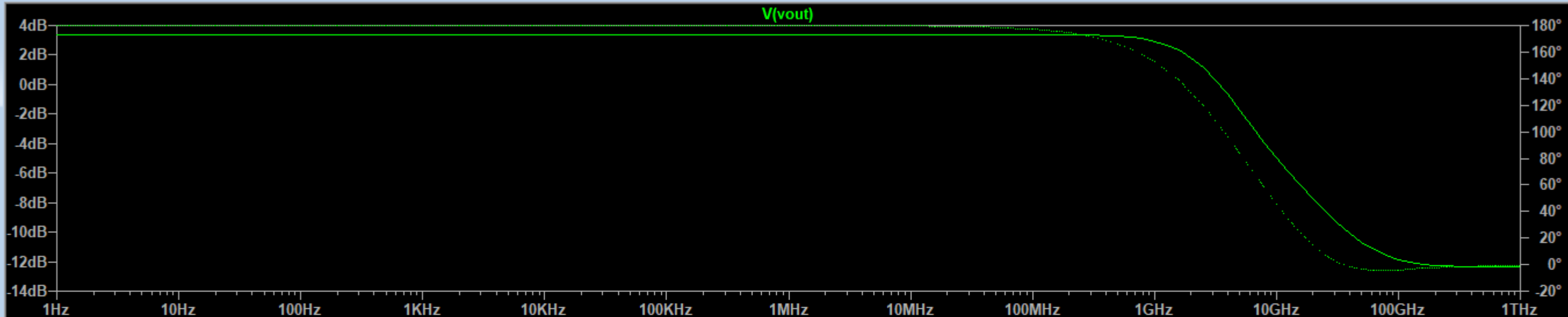
spiceparts

ERRORS

JOBS



ac_cascode_amp.spi

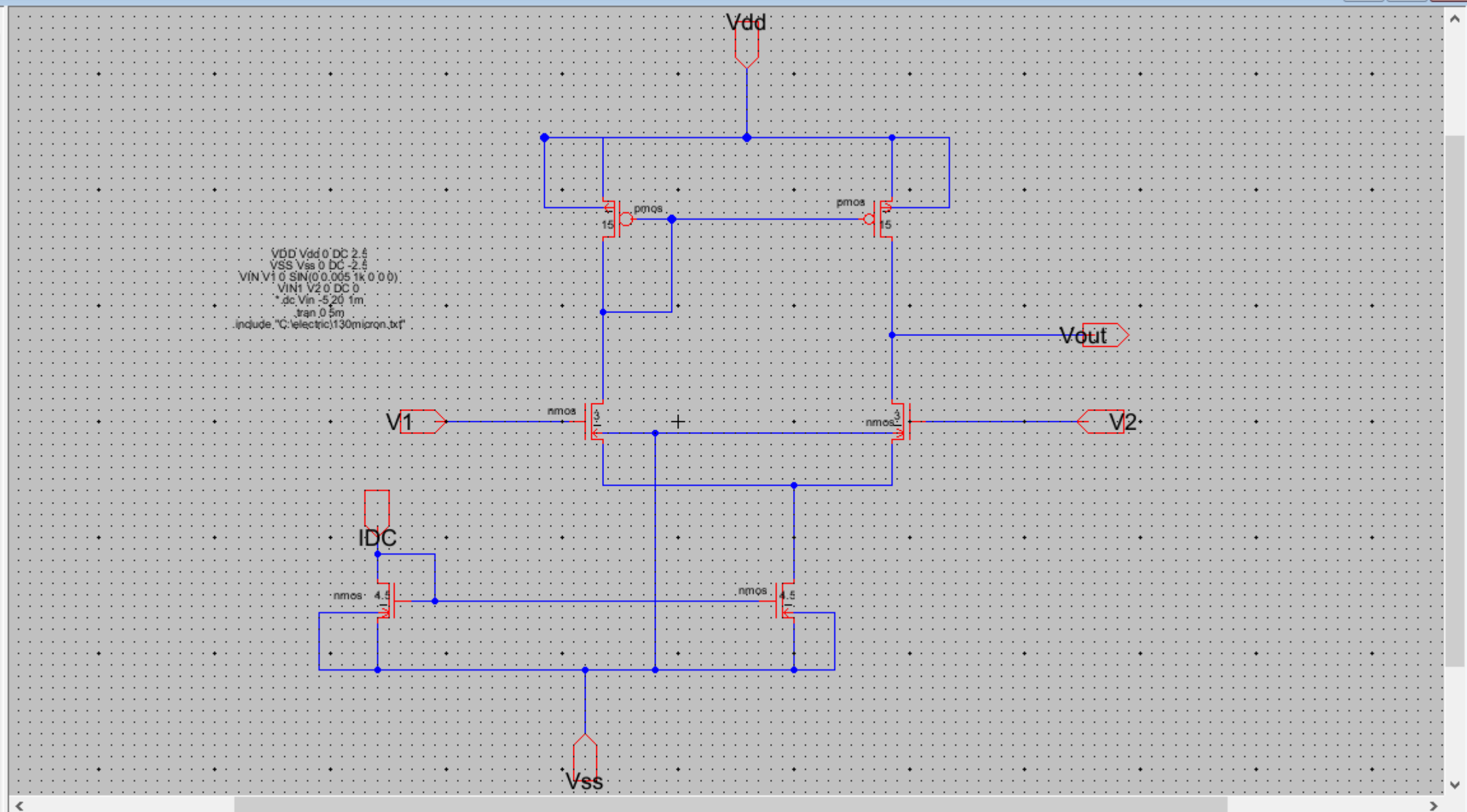
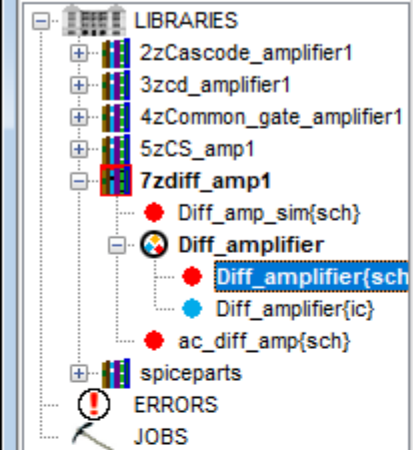


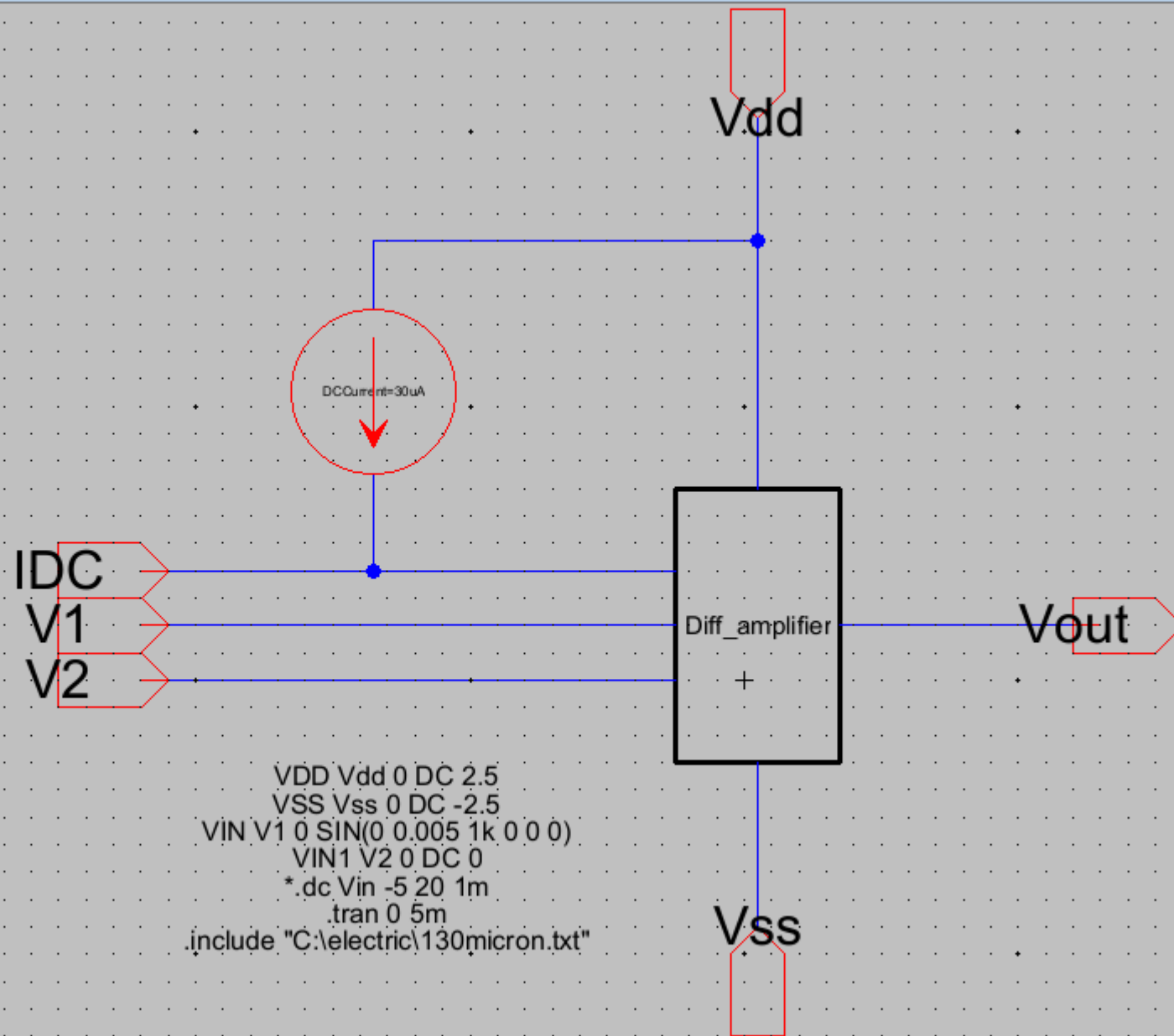
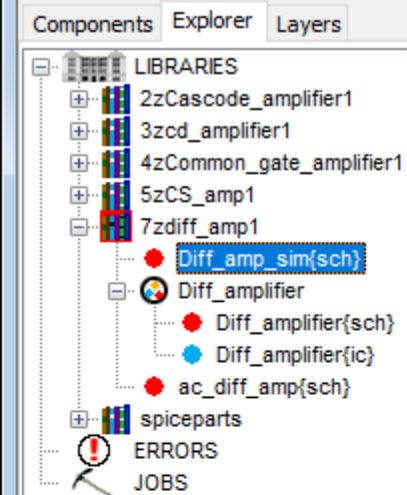
ac_cascode_amp.spi

```
*** TOP LEVEL CELL: ac_cascode_amp{sch}
Mnmos-4@0 Vout Vb net@11 Vss nmos L=0.4U W=10U
Mnmos-4@1 net@11 net@6 Vss Vss nmos L=0.4U W=10U
Rres@0 vdd Vout 10000
VACVtag@0 net@6 Vss DC=1V AC=2V SIN (1V 2V 1,000 1ns)

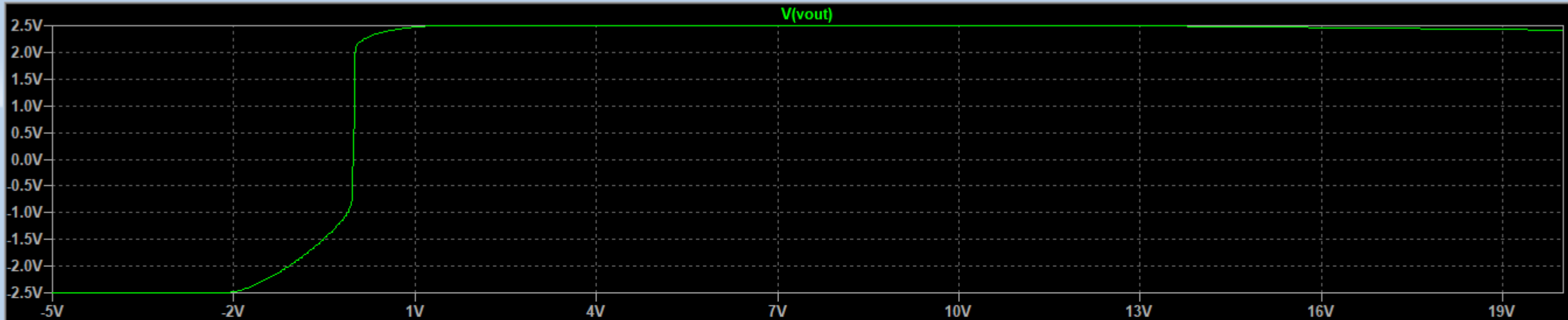
* Spice Code nodes in cell cell 'ac_cascode_amp{sch}'
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
VBias Vb 0 DC 2.5
.ac DEC 10 1 1T
.include "C:\130micron.txt"
.END
.END
```

Components Explorer Layers





Diff_amp_sim.spi



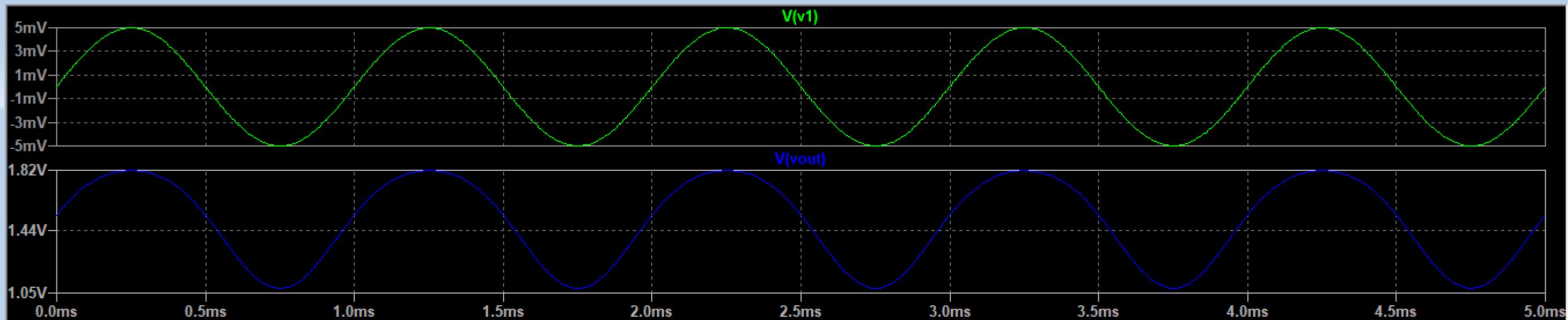
Diff_amp_sim.spi



```
*** TOP LEVEL CELL: Diff_amp_sim{sch}
IDCCurren@1 Vdd IDC DC 30uA
XDiff_amp@0 IDC V1 V2 Vout Vss Vdd Diff_amplifier

* Spice Code nodes in cell cell 'Diff_amp_sim{sch}'
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
VIN V1 0 SIN(0 0.005 1k 0 0 0)
VIN1 V2 0 DC 0
.dc Vin -5 20 1m
*.tran 0 5m
.include "C:\130micron.txt"
.END
```

Diff_amp_sim.spi



Diff_amp_sim.spi



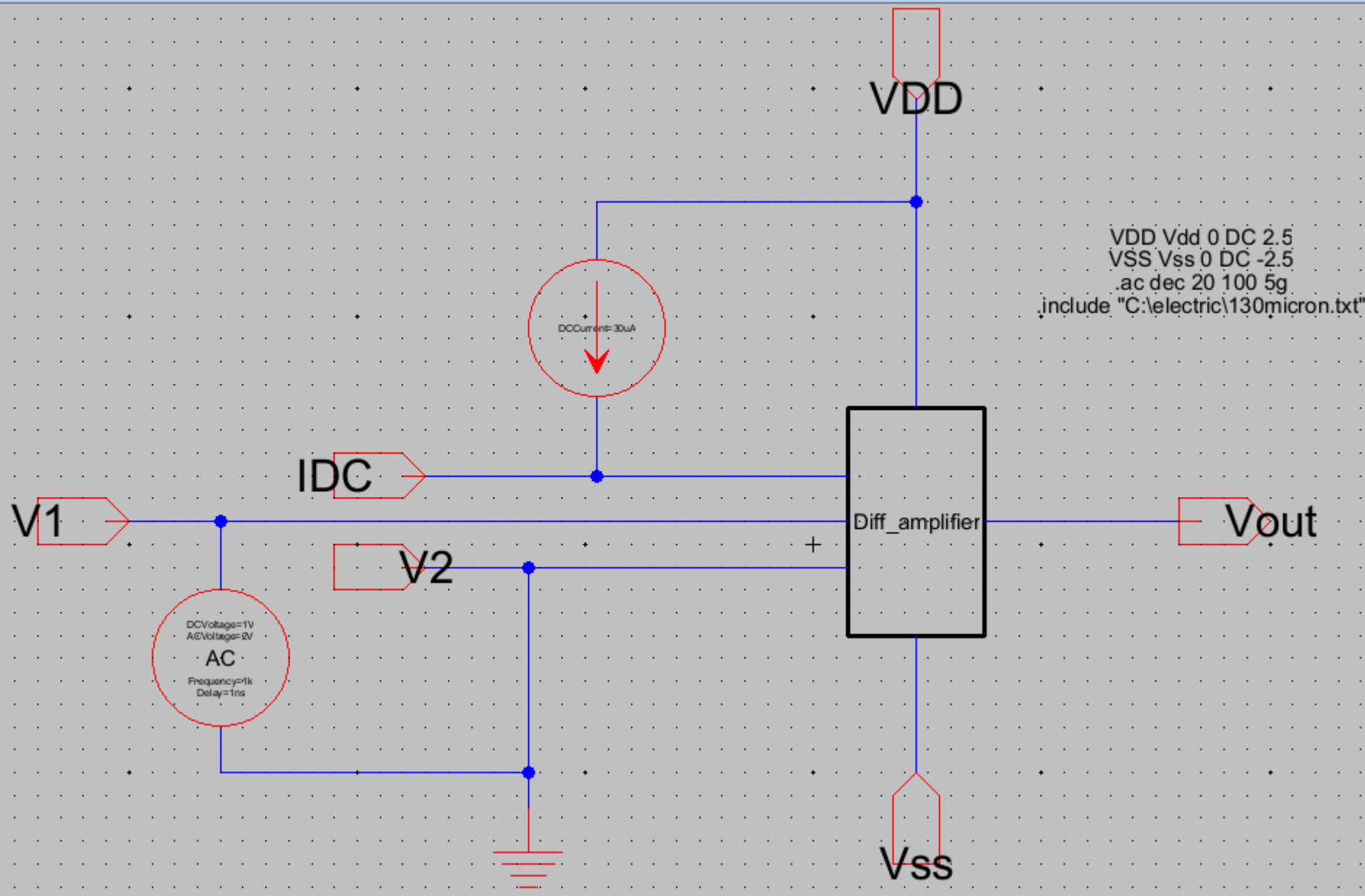
```
*** TOP LEVEL CELL: Diff_amp_sim{sch}
IDCCurren@1 Vdd IDC DC 30uA
XDiff_amp@0 IDC V1 V2 Vout Vss Vdd Diff_amplifier

* Spice Code nodes in cell cell 'Diff_amp_sim{sch}'
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
VIN V1 0 SIN(0 0.005 1k 0 0 0)
VIN1 V2 0 DC 0
*.dc Vin -5 20 1m
.tran 0 5m
.include "C:\130micron.txt"
.END
```

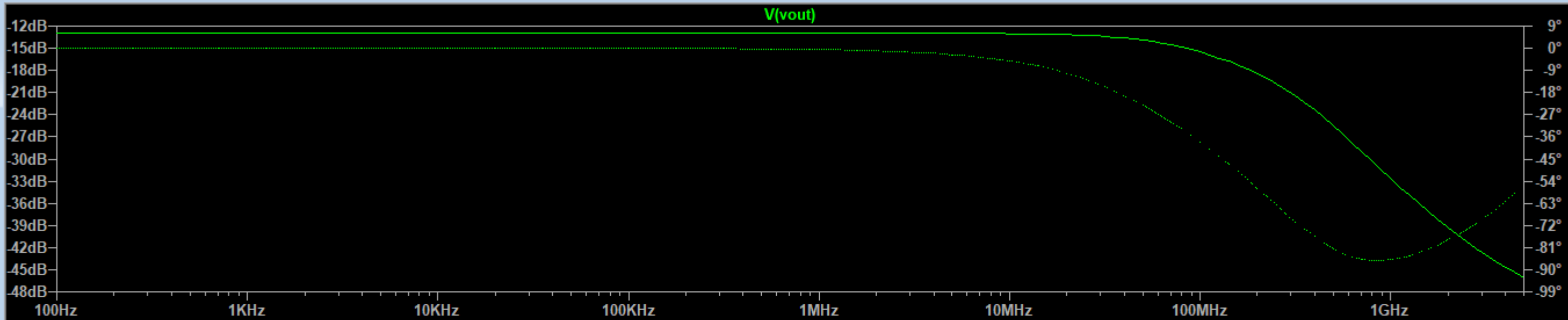

Components Explorer Layers

LIBRARIES

- 2zCascode_amplifier1
- 3zcd_amplifier1
- 4zCommon_gate_amplifier1
- 5zCS_amp1
- 7zdiff_amp1
 - Diff_amp_sim{sch}
 - Diff_amplifier
 - Diff_amplifier{sch}
 - Diff_amplifier{ic}
 - ac_diff_amp{sch}
- spiceparts
- ERRORS
- JOBS



ac_diff_amp.spi



ac_diff_amp.spi

```
.global gnd

*** TOP LEVEL CELL: ac_diff_amp{sch}
VACVoltage@0 V1 gnd DC=1V AC=2V SIN (1V 2V 1,000 1ns)
IDCCurrent@0 vdd IDC DC 30uA
XDiff_amp@0 IDC V1 gnd Vout Vss vdd Diff_amplifier

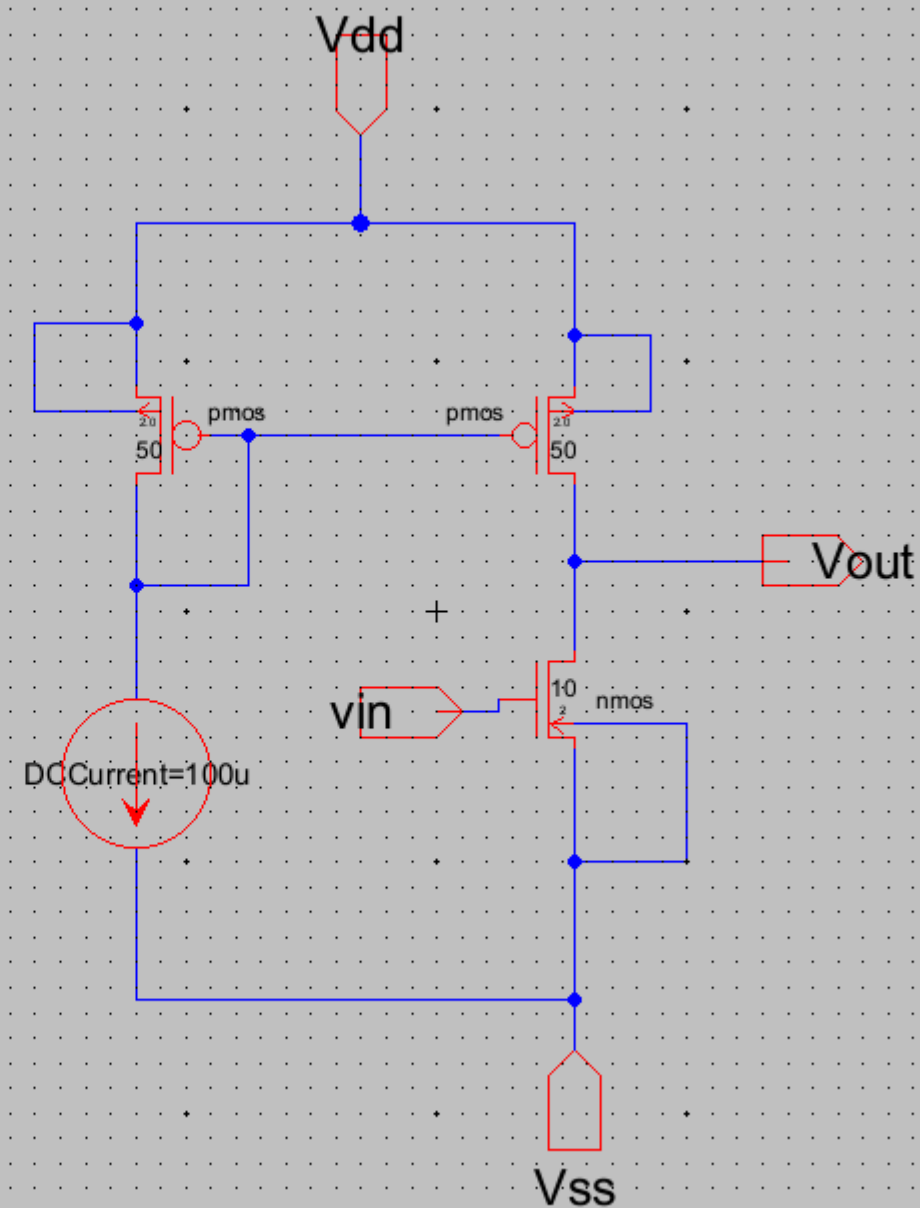
* Spice Code nodes in cell cell 'ac_diff_amp{sch}'
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
.ac dec 20 100 5g
.include "C:\130micron.txt"
.END
```

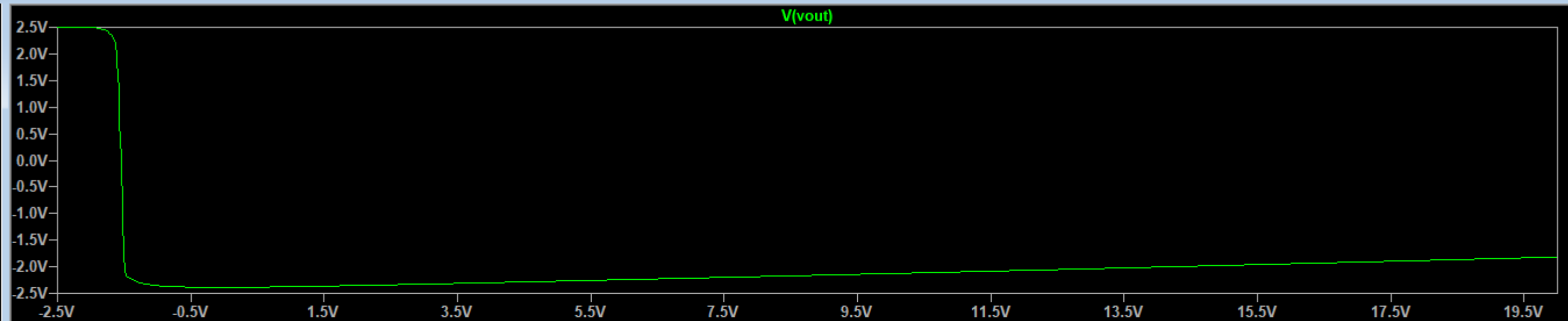
Components Explorer Layers

LIBRARIES

- 2zCascode_amplifier1
- 3zcd_amplifier1
- 4zCommon_gate_amplifier1
- 5zCS_amp1
- 6zCurrent_mirror1
- Current_mirror(sch)
- zCurrent_mirror1_ac(sc)
- 7zdiff_amp1
- spiceparts
- ERRORS
- JOBS

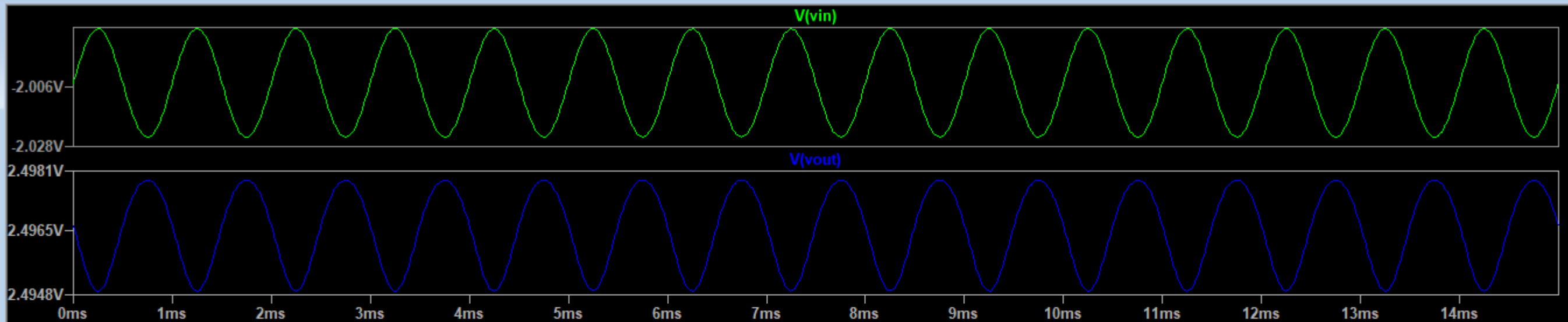
```
.VDD Vdd 0 DC 2.5  
.VSS Vss 0 DC -2.5  
.VIN vin 0 SIN(-2.0045 20m 1k 0 0 0)  
*VIN1 V2 0 DC 0  
.dc Vin -2.5 20 1m  
*tran 0 15m  
.include "C:\130micron.txt"
```





```
Mnmos-4@3 Vout vin Vss Vss nmos L=0.4U W=2U
Mpmos-4@1 Vout net@10 vdd vdd pmos L=0.4U W=10U
Mpmos-4@2 net@10 net@10 vdd vdd pmos L=0.4U W=10U
IDCCurren@1 net@10 Vss DC 100u
```

```
* Spice Code nodes in cell cell 'Current_mirror{sch}'
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
VIN vin 0 SIN(-2.0045 20m 1k 0 0 0)
*VIN1 V2 0 DC 0
.dc Vin -2.5 20 1m
*.tran 0 15m
.include "C:\130micron.txt"
.END
```



```
Mnmos-4@3 Vout vin Vss Vss nmos L=0.4U W=2U
Mpmos-4@1 Vout net@10 vdd vdd pmos L=0.4U W=10U
Mpmos-4@2 net@10 net@10 vdd vdd pmos L=0.4U W=10U
IDCCurren@1 net@10 Vss DC 100u
```

```
* Spice Code nodes in cell cell 'Current_mirror{sch}'
```

```
VDD Vdd 0 DC 2.5
```

```
VSS Vss 0 DC -2.5
```

```
VIN vin 0 SIN(-2.0045 20m 1k 0 0 0)
```

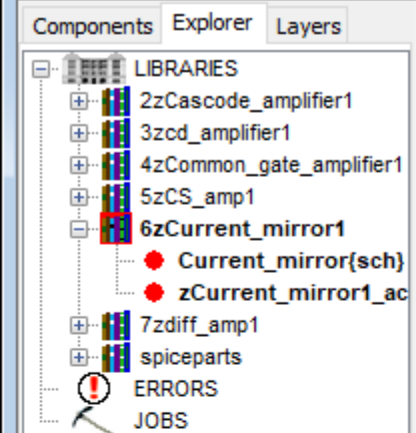
```
*VIN1 V2 0 DC 0
```

```
*.dc Vin -2.5 20 1m
```

```
.tran 0 15m
```

```
.include "C:\130micron.txt"
```

```
.END
```

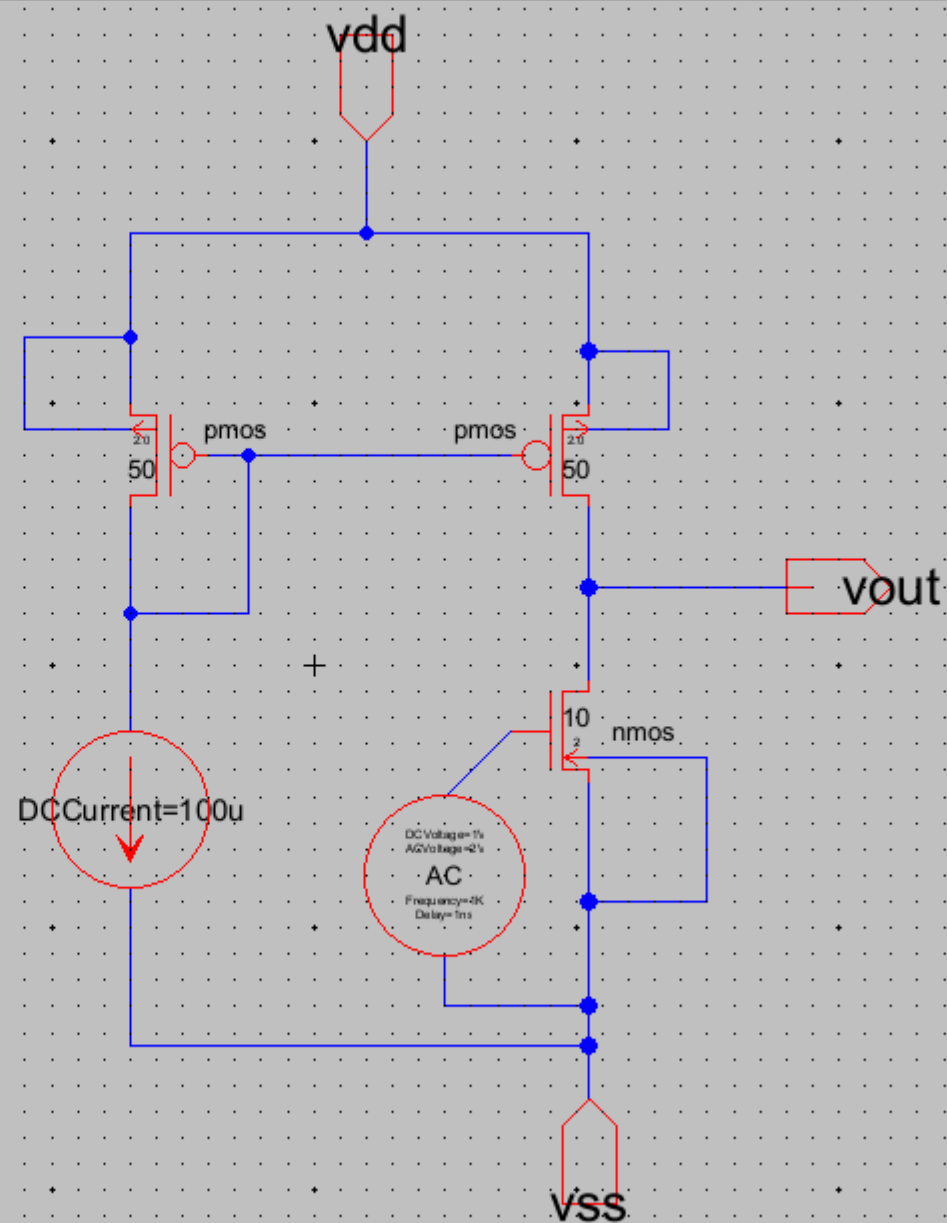


VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
ac dec 20 100 100g
.include "C:\130micron.txt"

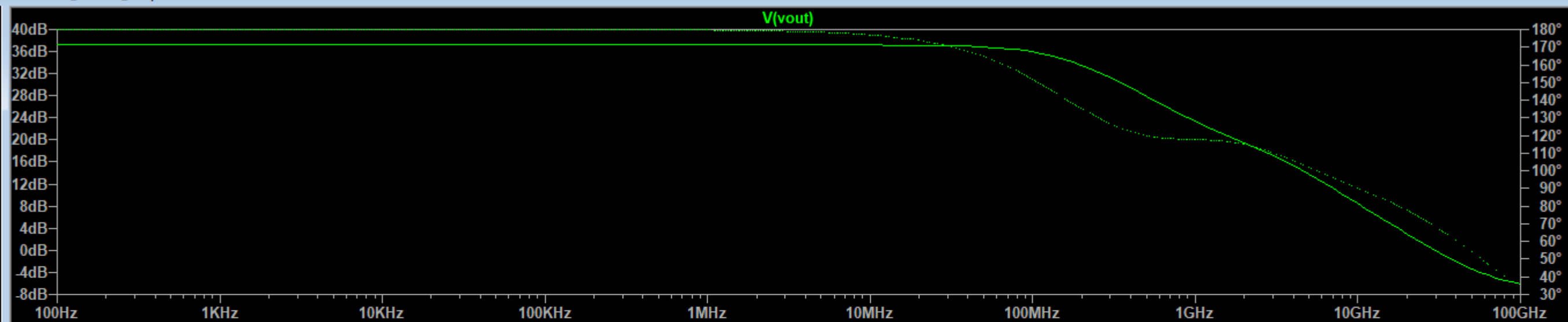
DCCurrent=100u

AC

Frequency=2K
Delay=1ns



zCurrent_mirror1_ac.spi



zCurrent_mirror1_ac.spi



```
*** TOP LEVEL CELL: zCurrent_mirror1_ac{sch}
Mnmos-4@0 vout net@32 vss vss nmos L=0.4U W=2U
Mpmos-4@0 vout net@0 vdd vdd pmos L=0.4U W=10U
Mpmos-4@1 net@0 net@0 vdd vdd pmos L=0.4U W=10U
VACVoltage@0 net@32 vss DC=1V AC=2V SIN (1V 2V 1,000 1ns)
IDCCurren@0 net@0 vss DC 100u

* Spice Code nodes in cell cell 'zCurrent_mirror1_ac{sch}'
VDD Vdd 0 DC 2.5
VSS Vss 0 DC -2.5
.ac dec 20 100 100g
.include "C:\130micron.txt"
.END
```

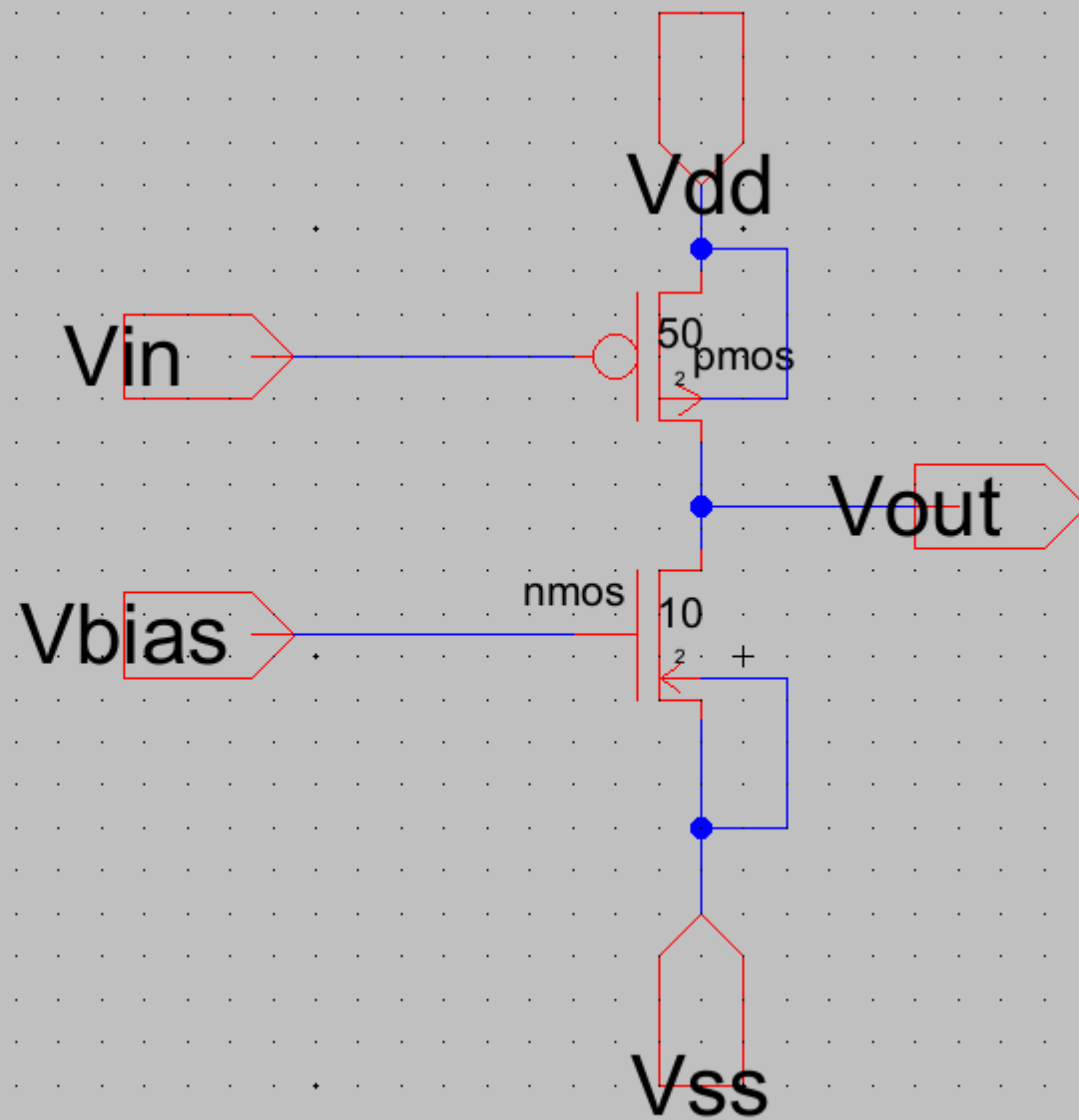
Components Explorer Layers

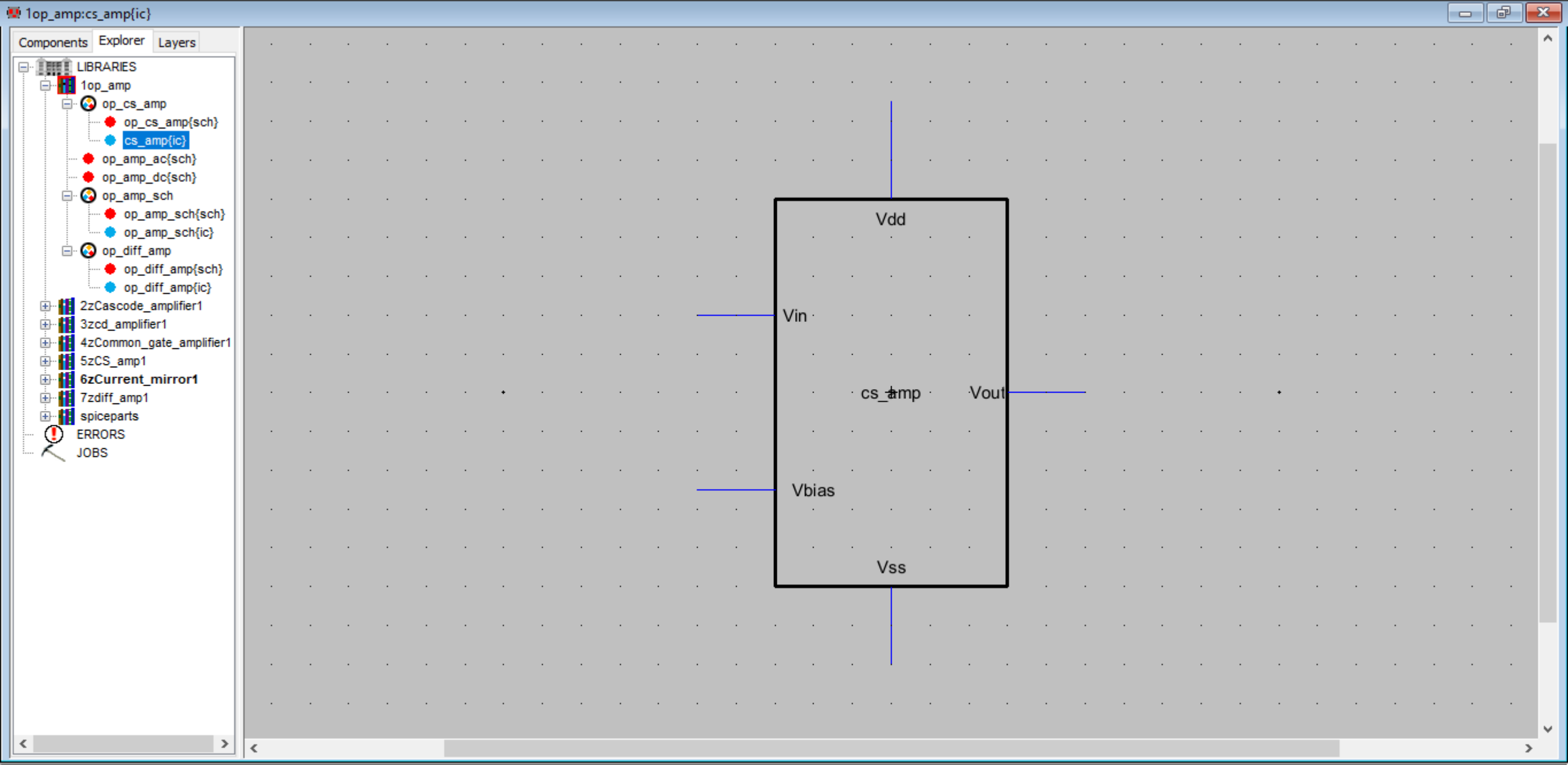
LIBRARIES

- 1op_amp
 - op_cs_amp
 - op_cs_amp{sch}
 - cs_amp{ic}
 - op_amp_ac{sch}
 - op_amp_dc{sch}
- op_amp_sch
 - op_amp_sch{sch}
 - op_amp_sch{ic}
- op_diff_amp
 - op_diff_amp{sch}
 - op_diff_amp{ic}

- 2zCascode_amplifier1
- 3zcd_amplifier1
- 4zCommon_gate_amplifier1
- 5zCS_amp1
- 6zCurrent_mirror1
- 7zdiff_amp1
- spiceparts

- ERRORS
- JOBS





Components Explorer Layers

LIBRARIES

1op_amp

op_cs_amp

op_cs_amp{sch}

cs_amp{ic}

op_amp_ac{sch}

op_amp_dc{sch}

op_amp_sch

op_amp_sch{sch}

op_amp_sch{ic}

op_diff_amp

op_diff_amp{sch}

op_diff_amp{ic}

2zCascode_amplifier1

3zcd_amplifier1

4zCommon_gate_amplifier1

5zCS_amp1

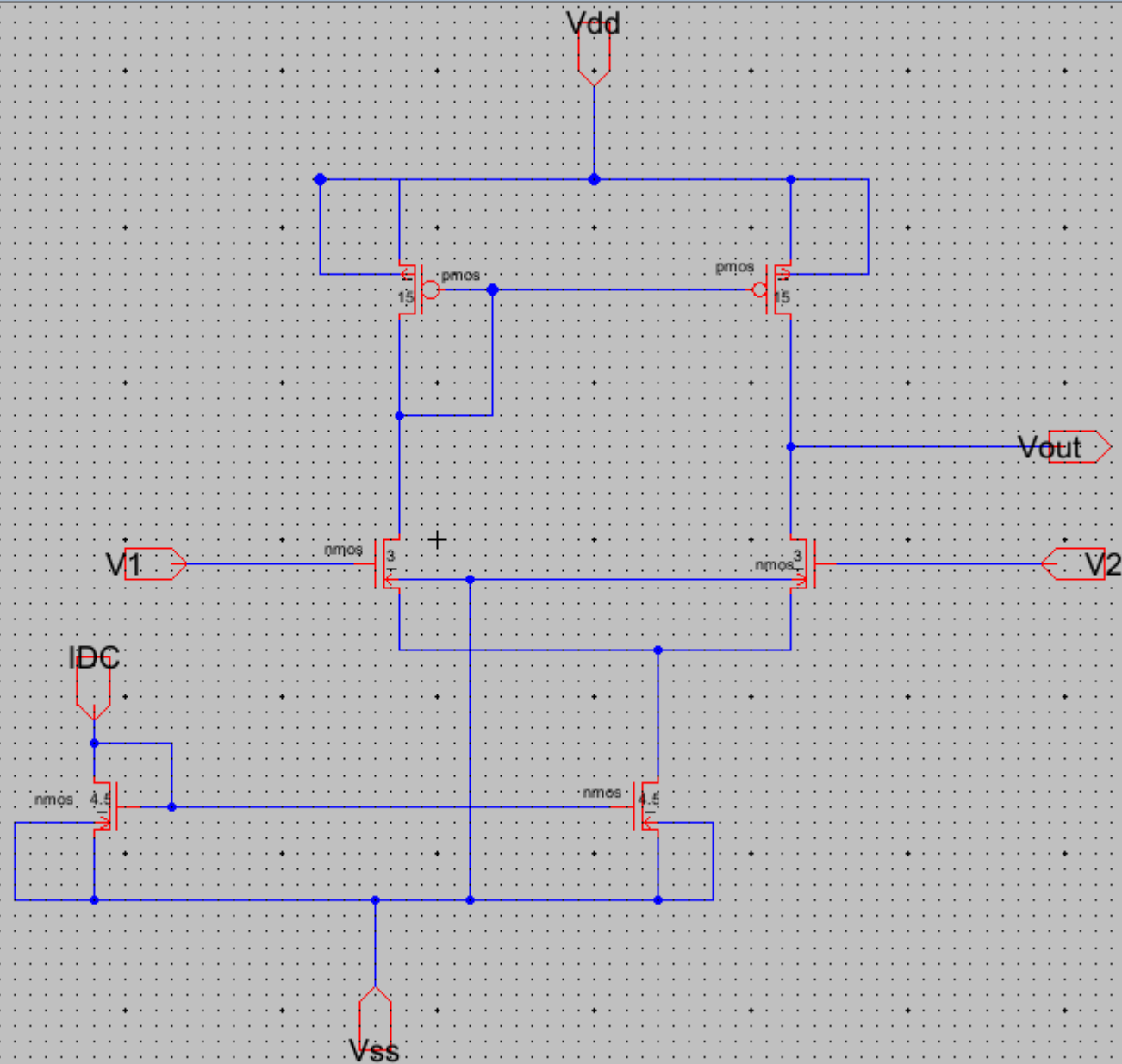
6zCurrent_mirror1

7zdiff_amp1

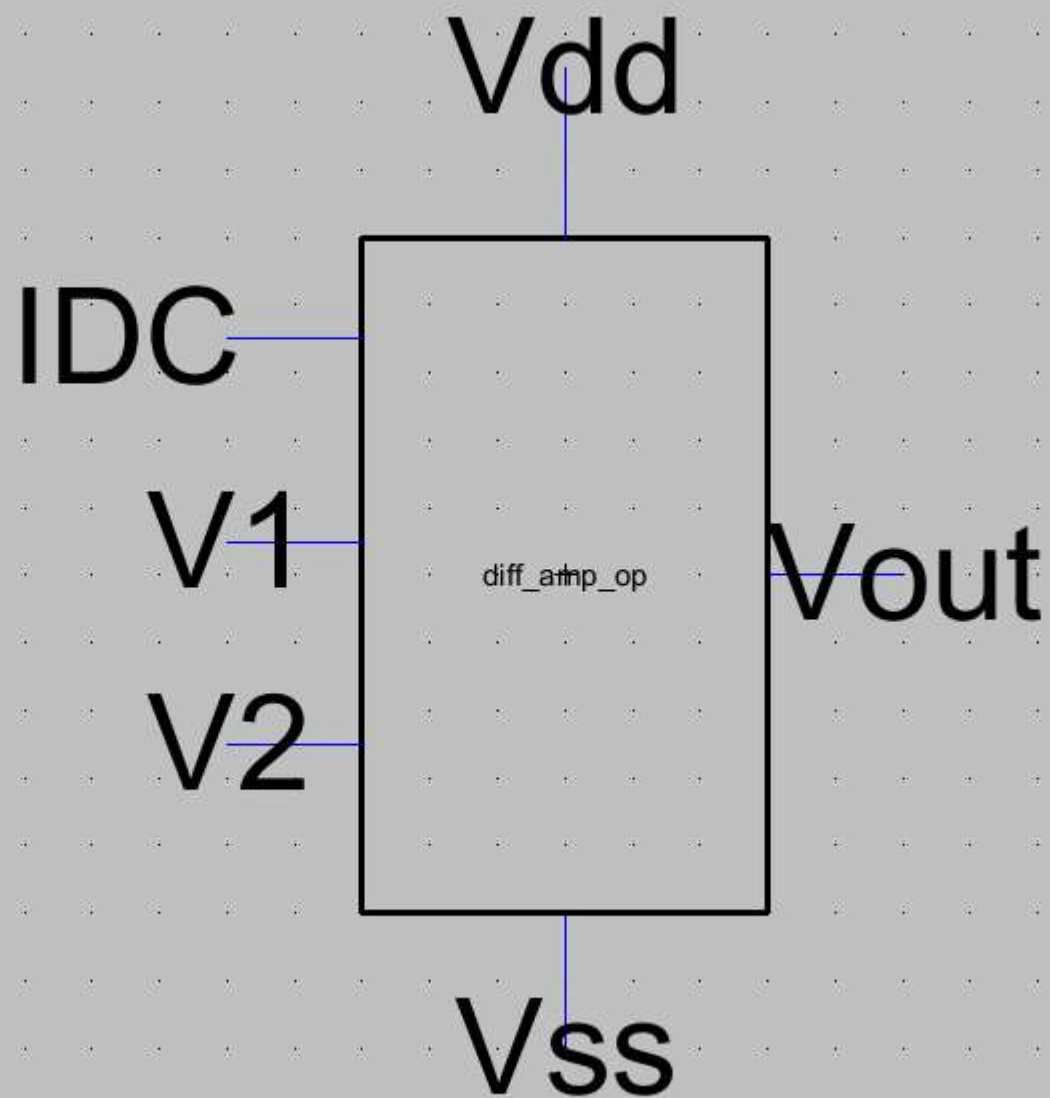
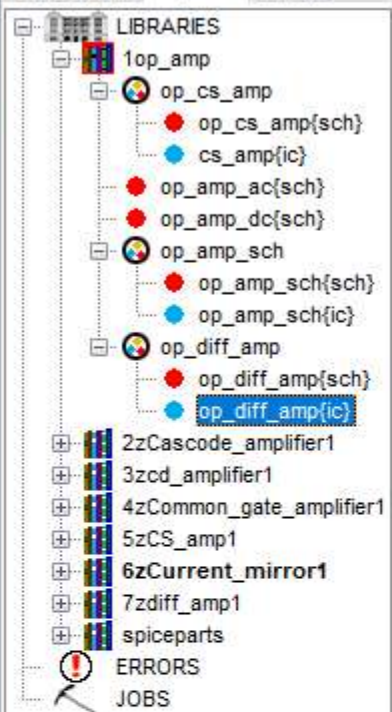
spiceparts

ERRORS

JOBS



Components Explorer Layers



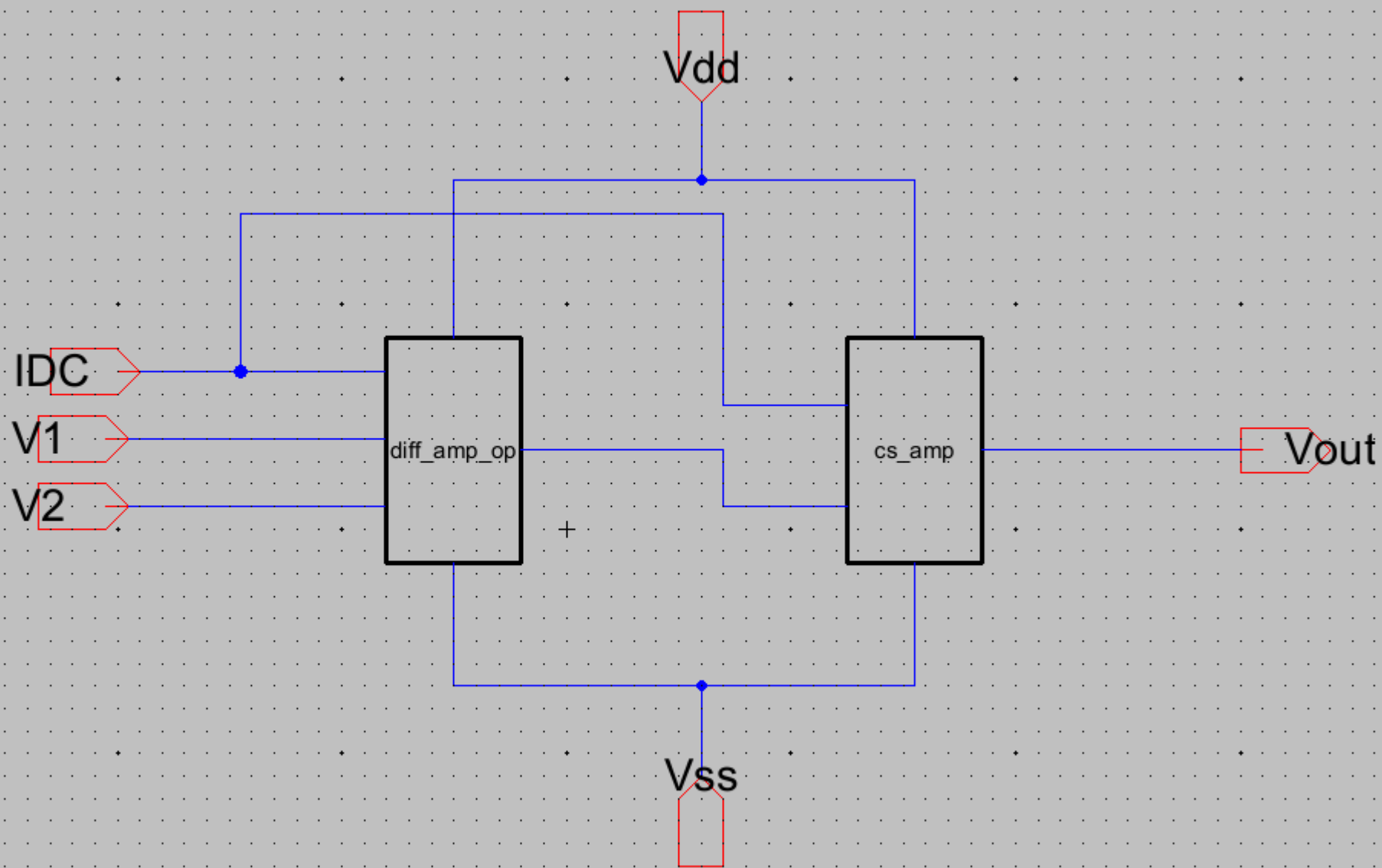
Components Explorer Layers

LIBRARIES

1op_amp

- op_cs_amp
 - op_cs_amp{sch}
 - cs_amp{ic}
- op_amp_ac{sch}
- op_amp_dc{sch}
- op_amp_sch
 - op_amp_sch{sch}
 - op_amp_sch{ic}
- op_diff_amp
 - op_diff_amp{sch}
 - op_diff_amp{ic}

- 2zCascode_amplifier1
- 3zcd_amplifier1
- 4zCommon_gate_amplifier1
- 5zCS_amp1
- 6zCurrent_mirror1
- 7zdiff_amp1
- spiceparts

 ERRORS
 JOBS

Components Explorer Layers

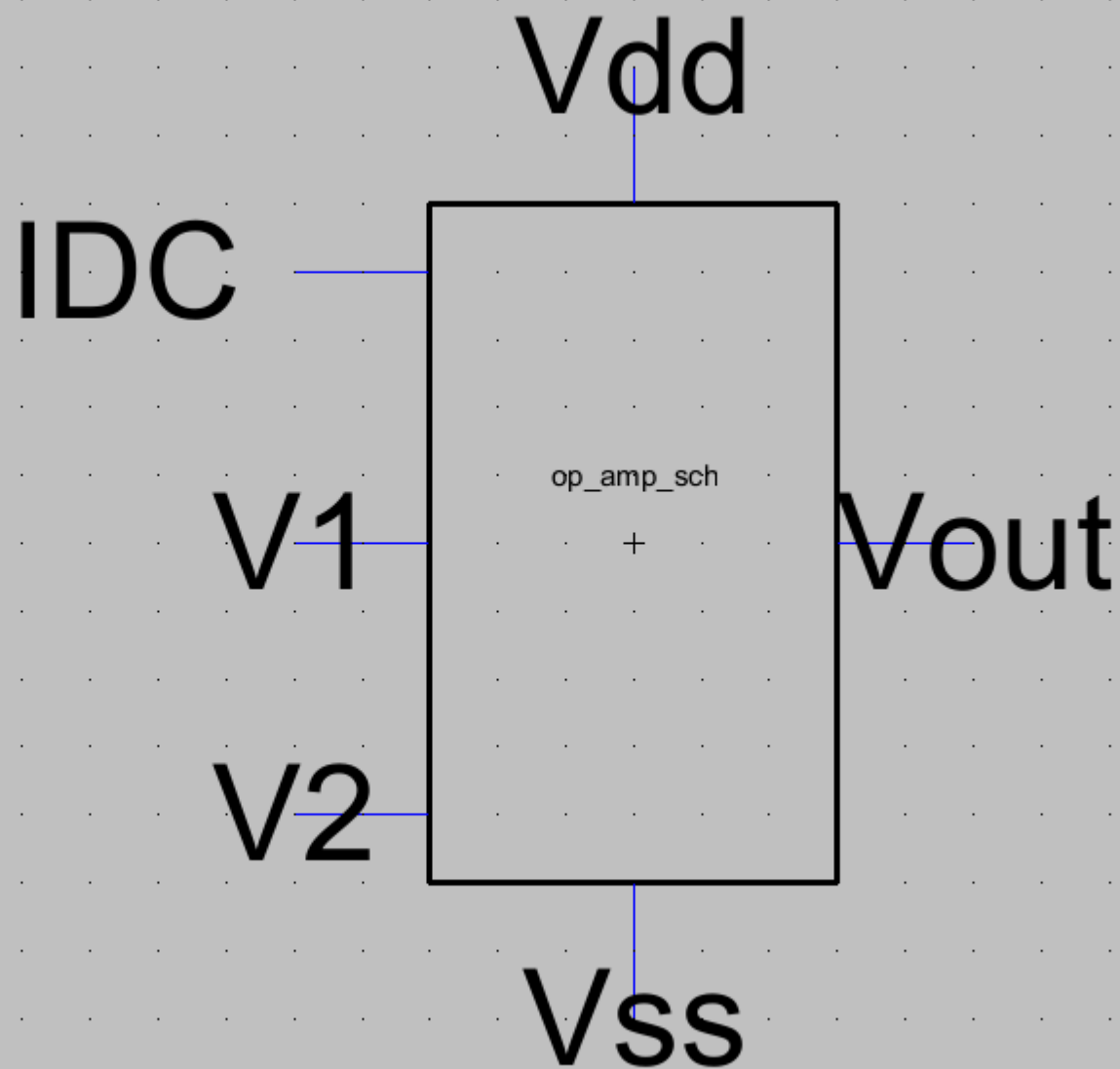
LIBRARIES

1op_amp

- op_cs_amp
 - op_cs_amp{sch}
 - cs_amp{ic}
- op_amp_ac{sch}
- op_amp_dc{sch}
- op_amp_sch
 - op_amp_sch{sch}
 - op_amp_sch{ic}
- op_diff_amp
 - op_diff_amp{sch}
 - op_diff_amp{ic}

- 2zCascode_amplifier1
- 3zcd_amplifier1
- 4zCommon_gate_amplifier1
- 5zCS_amp1
- 6zCurrent_mirror1
- 7zdiff_amp1
- spiceparts

- ERRORS
- JOBS



Components Explorer Layers

LIBRARIES

1op_amp

op_cs_amp

op_cs_amp{sch}

cs_amp{ic}

op_amp_ac{sch}

op_amp_dc{sch}

op_amp_sch

op_amp_sch{sch}

op_amp_sch{ic}

op_diff_amp

op_diff_amp{sch}

op_diff_amp{ic}

2zCascode_amplifier1

3zcd_amplifier1

4zCommon_gate_amplifier1

5zCS_amp1

6zCurrent_mirror1

7zdiff_amp1

spiceparts

ERRORS

JOBS

```
.Vdd Vdd 0 dc 2.5
.Vss Vss 0 DC -2.5
.Vin V1 0 sin(0 0.002 1k)
.Vin1 V2 0 DC 0
.dc Vin -2.5 20 1m
*.tran 0 5m
.include "C:\130micron.txt"
```

IDC

DCCurrent=30uA

V1

V2

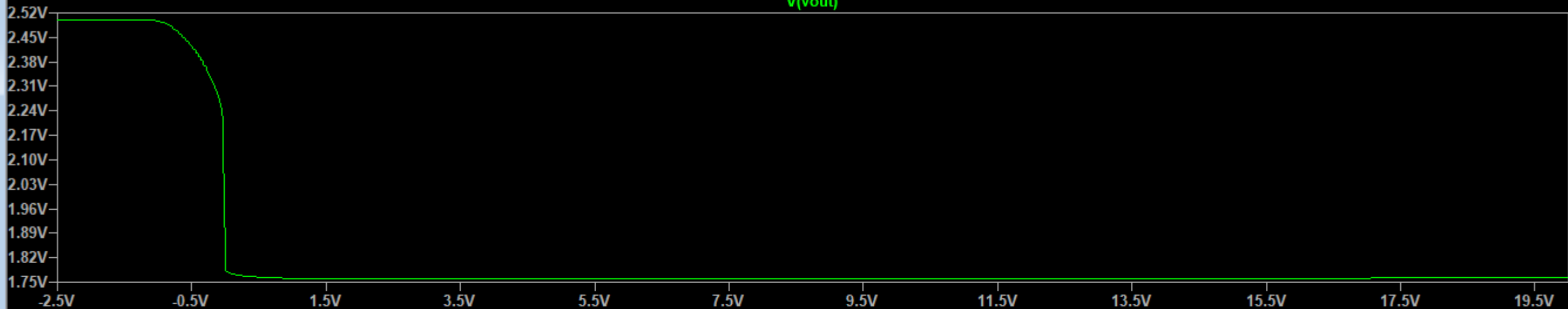
Vdd

op_amp_sch

Vout

Vss

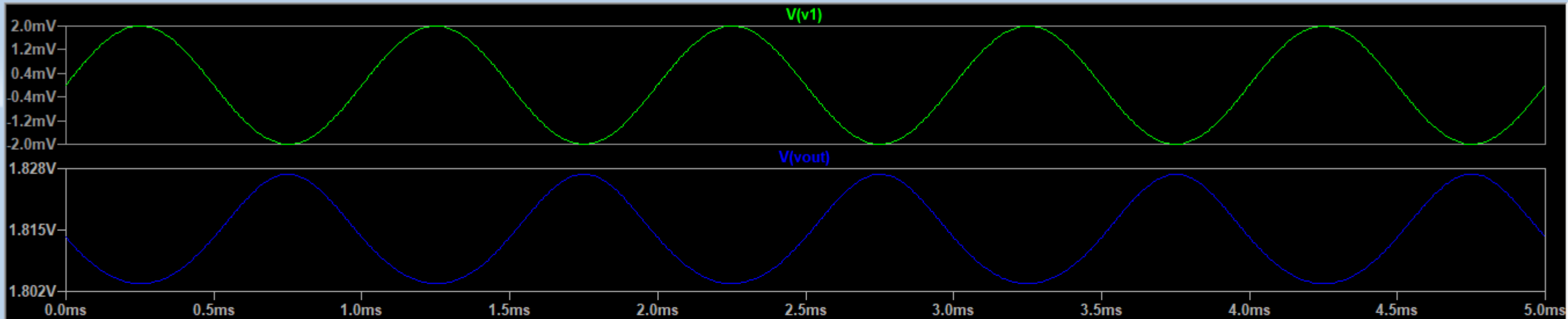
V(vout)



```
*** TOP LEVEL CELL: op_amp_dc{sch}  
IDCCurren@0 vdd IDC DC 30uA  
Xop_amp_s@1 IDC V1 V2 Vout Vss vdd op_amp_sch
```

```
* Spice Code nodes in cell cell 'op_amp_dc{sch}'  
Vdd Vdd 0 dc 2.5  
Vss Vss 0 DC -2.5  
Vin V1 0 sin(0 0.002 1k)  
Vin1 V2 0 DC 0  
.dc Vin -2.5 20 1m  
*.tran 0 5m  
.include "C:\130micron.txt"  
.END
```

op_amp_dc.spi



op_amp_dc.spi



```
*** TOP LEVEL CELL: op_amp_dc{sch}
IDCCurren@0 vdd IDC DC 30uA
Xop_amp_s@1 IDC V1 V2 Vout Vss vdd op_amp_sch

* Spice Code nodes in cell cell 'op_amp_dc{sch}'
Vdd Vdd 0 dc 2.5
Vss Vss 0 DC -2.5
Vin V1 0 sin(0 0.002 1k)
Vin1 V2 0 DC 0
*.dc Vin -2.5 20 1m
|.tran 0 5m
.include "C:\130micron.txt"
.END
```


Components Explorer Layers

LIBRARIES

1op_amp

op_cs_amp

op_cs_amp{sch}

cs_amp{ic}

op_amp_ac{sch}

op_amp_dc{sch}

op_amp_sch

op_amp_sch{sch}

op_amp_sch{ic}

op_diff_amp

op_diff_amp{sch}

op_diff_amp{ic}

2zCascode_amplifier1

3zcd_amplifier1

4zCommon_gate_amplifier1

5zCS_amp1

6zCurrent_mirror1

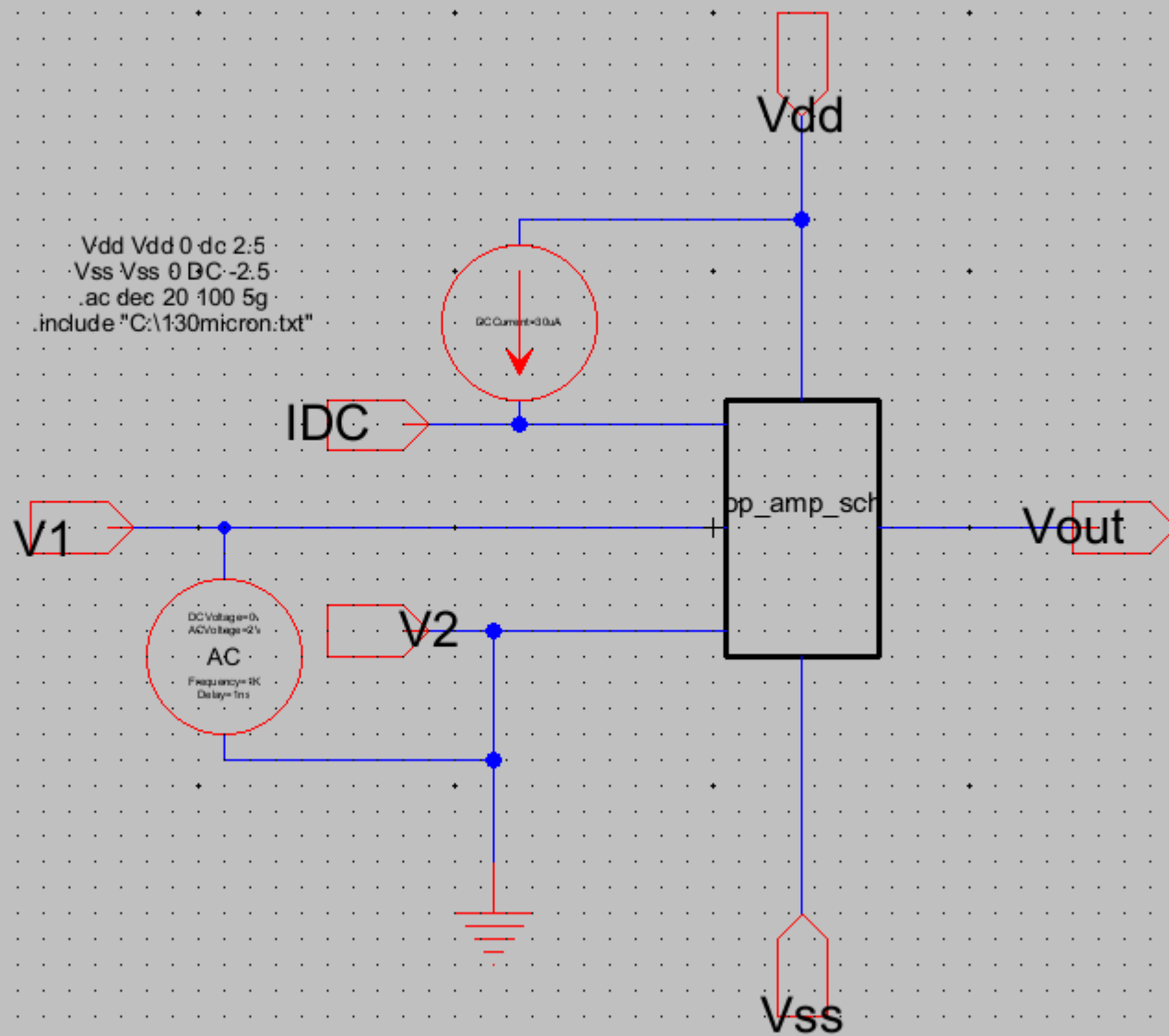
7zdiff_amp1

spiceparts

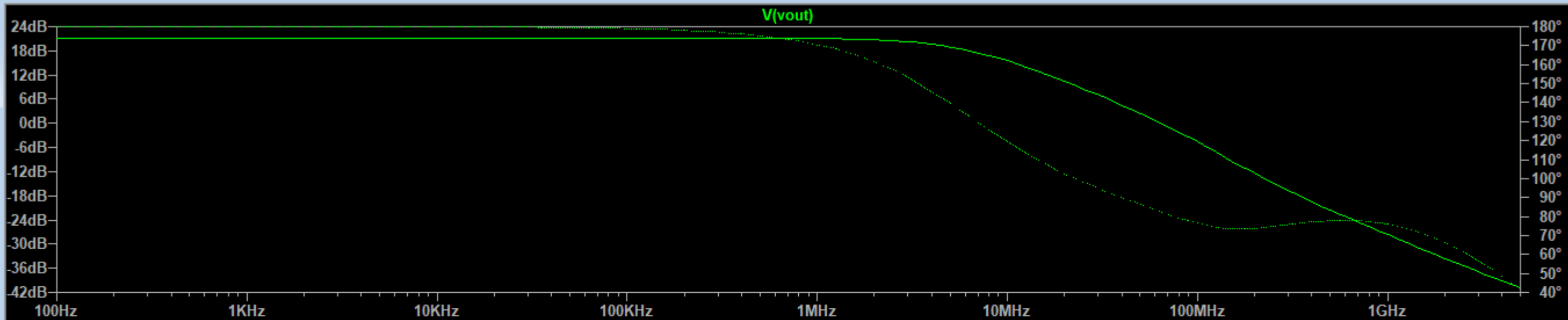
ERRORS

JOBS

```
Vdd Vdd 0 dc 2.5
Vss Vss 0 DC -2.5
.ac dec 20 100 5g
.include "C:\130micron.txt"
```



op_amp_ac.spi



op_amp_ac.spi

```
.global gnd

*** TOP LEVEL CELL: op_amp_ac{sch}
VACVtag@0 V1 gnd DC=0v AC=2V SIN (0v 2V 1,000 1ns)
IDCCurren@0 vdd IDC DC 30uA
Xop_amp_s@0 IDC V1 gnd Vout Vss vdd op_amp_sch

* Spice Code nodes in cell cell 'op_amp_ac{sch}'
Vdd Vdd 0 dc 2.5
Vss Vss 0 DC -2.5
.ac dec 20 100 5g
.include "C:\130micron.txt"
.END
```